Matrix-Aware Optimizers: A Survey

Kyunghun Nam

Kentech

baptist0214@gmail.com

8 October 2026

Abstract

Matrix-aware optimizers can reduce the number of steps needed to train a neural network, yet their additional matrix operations, state, and communication may offset that gain. This survey examines the trade-off through Shampoo, SOAP, and Muon, following the information each method retains to the update it computes and the cost of computing it. We relate structured preconditioning to polar updates, identify when these maps coincide, and explain why a factorized basis can support nonseparable adaptive scales. A stochastic-quadratic model then separates finite-time contraction from noise accumulation, showing why curvature and noise orientation must be considered together. We compare convergence results with their recursions, assumptions, and stationarity measures intact, before examining finite matrix computations, state transport under basis changes, and distributed updates. Published experiments illustrate where iteration efficiency, numerical accuracy, and runtime lead to different conclusions. The resulting account identifies what is understood about these methods and what is still needed to predict whether matrix structure reduces the time to a target quality.

Keywords: adaptive optimization; Shampoo; Muon; Kronecker structure; convergence theory; matrix functions; distributed optimization

Contents

1 Introduction . . . 2
2 Mathematical foundations . . . 3
3 Structured adaptive preconditioning . . . 6
4 Spectral and adaptive spectral updates . . . 8
5 Algebraic connections and optimization dynamics . . . 12
6 Comparing theoretical guarantees . . . 15
7 Inexact matrix computation . . . 21
8 Distributed realizations and structural fidelity . . . 25
9 Published empirical evidence and its interpretation . . . 28
10 Open problems and predictive theory . . . 30
11 Conclusion . . . 32
References . . . 33
A Proof details and auxiliary identities . . . 38
B Theorem statements and assumptions . . . 40
C Reference specifications and historical context . . . 44
D Empirical records . . . 48

1 Introduction

Fewer optimization steps do not necessarily mean cheaper training. Matrix-aware methods use the row–column structure of parameter blocks to shape their updates, but the resulting matrix operations, optimizer state, and communication all have a cost. Their practical value therefore depends on the quality reached within a training budget, rather than on iteration counts alone.

AdamW provides a useful baseline: it combines coordinatewise adaptation with decoupled weight decay, retaining first moments and elementwise second moments [1, 2]. Shampoo, SOAP, and Muon depart from this construction in different ways. Shampoo accumulates historical tensor-mode Gram statistics to build its preconditioners [3, 4]. SOAP uses a factorized basis obtained from such statistics, but learns coordinatewise scales within that basis [5]. Muon instead transforms the singular values of a momentum-based matrix through an approximate polar map [6, 7]. These are different uses of matrix structure, not interchangeable implementations of one update.

Published language-model studies report improvements over AdamW in iteration counts and, under particular training recipes, in wall-clock time or training FLOPs [5, 8]. Interpreting those results requires following the full computation. A better estimate may be expensive to maintain; a more accurate matrix transform may not improve the training trajectory; and a distributed implementation may change the update it was intended to accelerate.

We organize the comparison around this path from information to computation. For each method, we distinguish its statistical state, update geometry, numerical computation, and distributed representation. The separation lets us ask whether a proposed change improves an estimator, defines a different update, or computes the same update more efficiently. It also clarifies several recurring ambiguities: gradient second moments need not represent curvature, a finite polynomial need not approximate polar uniformly, and a factorized basis need not define a Kronecker-product operator.

The mathematical development begins with a common update specification and the conditions under which different maps coincide (sections 4.5 and 5). A frozen stochastic-quadratic model then shows how a preconditioner affects both contraction of the initial error and accumulation of noise. We use this distinction when comparing convergence guarantees in section 6, and carry it into the numerical and distributed analyses in sections 7 and 8. The propositions provide derivations that support the comparison; the survey introduces neither a new optimizer nor a new convergence rate.

Related work and scope. Norm-based and adapted-polar interpretations, KL-based factor fitting, and rotation-based optimization supply the main conceptual foundations [7, 9–13]. Tables 8 and 9 place these connections in their historical context. We select sources for distinct estimation objectives, update maps, proof techniques, numerical methods, or distributed implementations, rather than aiming for an exhaustive list of optimizers. Broader surveys cover more families [14]; quasi-Newton, natural-gradient, decentralized, and low-memory adaptive methods are outside our main scope. K-FAC and Lie-group preconditioning appear as related approaches [15, 16]. The appendices collect proof details, theorem assumptions, method specifications, and the conditions of the cited experiments.

2 Mathematical foundations

2.1 Matrix geometry and stochastic information

Let $W \in \mathbb{R}^{m\times n}$ be a parameter block, G_t its stochastic gradient, and F an objective that may couple many blocks. We use the Frobenius pairing $\langle A, B\rangle = \operatorname{tr}(A^\top B)$ and column-major vectorization, so that

$$\operatorname{vec}(AGB) = (B^\top \otimes A)\operatorname{vec}(G) = (B^\top \otimes A)g. \tag{1}$$

For $A \succeq 0$, matrix powers and pseudoinverse powers are defined spectrally. Negative powers require positive definiteness unless a support-restricted identity is stated; in that case, the inverse power is set to zero on the nullspace.

For a rank-r matrix $Z = U_r\Sigma_r V_r^\top$, the canonical polar factor is

$$\operatorname{polar}(Z) = U_r V_r^\top, \qquad \operatorname{polar}(0) = 0. \tag{2}$$

This factor is a partial isometry: $\|\operatorname{polar}(Z)\|_F^2 = r$, and its spectral norm is one for $Z \neq 0$. We do not complete a rank-deficient input's nullspace to an arbitrary orthogonal factor. The norm relations used below are

$$\|Z\|_F \le \|Z\|_* \le \sqrt{r}\,\|Z\|_F\,, \qquad |\langle A, B\rangle| \le \|A\|_*\,\|B\|_{\mathrm{op}}\,. \tag{3}$$

The duality between nuclear and spectral norms will determine the polar descent direction and the stationarity measure associated with it.

To specify the stochastic information available at an update, let $\mathcal{F}_{t-1}$ contain the history before the current sample, with W_t measurable with respect to that history. A conventional unbiased oracle satisfies

$$\mathbb{E}[G_t \mid \mathcal{F}_{t-1}] = \nabla F(W_t), \qquad N_t = G_t - \nabla F(W_t). \tag{4}$$

Unbiased gradients do not, by themselves, give unbiased preconditioned noise. When P_t is formed from G_t, it is generally not $\mathcal{F}_{t-1}$-measurable, and $\mathbb{E}[P_t \operatorname{vec}(N_t) \mid \mathcal{F}_{t-1}]$ need not vanish. A convergence proof must control this dependence or use a suitably delayed construction.

2.2 Second moment, covariance, Fisher, and Hessian

At a fixed parameter, write $g = \operatorname{vec}(G)$ and $\mu = \mathbb{E}[g]$. The uncentered second moment C contains both the mean gradient and the centered noise covariance Σ:

$$C = \mathbb{E}[gg^\top] = \Sigma + \mu\mu^\top, \qquad \Sigma = \operatorname{Cov}(g). \tag{5}$$

For a mean of B conditionally independent example gradients, this becomes

$$C_B = \mu\mu^\top + \frac{\Sigma_{\mathrm{sample}}}{B}. \tag{6}$$

A Gram statistic of the minibatch mean therefore changes the relative weight of mean and noise as the batch size changes. It is not the same statistic as an average of per-example Gram matrices.

Curvature is a separate object. The Hessian $H = \nabla^2 F$ describes local curvature and can be indefinite, whereas a model Fisher averages score outer products under the model distribution.

Identifying the Fisher with a Hessian requires a likelihood model and regularity conditions; identifying it with a gradient second moment also depends on the sampling distribution. K-FAC uses model structure to approximate curvature or Fisher information [15]. Shampoo accumulates gradient Gram matrices [3], whose interpretation as covariance estimates requires additional assumptions.

The population mode contractions

$$L = \mathbb{E}[GG^\top], \qquad R = \mathbb{E}[G^\top G] \tag{7}$$

are partial traces of C. They retain row–column information without determining the full $mn \times mn$ second-moment matrix: distinct matrices C can have the same contractions and different cross-mode correlations.

2.3 AdamW as a baseline and an algorithmic specification

With elementwise multiplication and division understood, the AdamW template is

$$\begin{aligned} M_t &= \beta_1 M_{t-1} + (1-\beta_1) G_t, \\ V_t &= \beta_2 V_{t-1} + (1-\beta_2) G_t^{\odot 2}, \\ W_{t+1} &= (1-\eta_t\lambda_t) W_t - \eta_t \frac{\widehat{M}_t}{\sqrt{\widehat{V}_t} + \varepsilon}. \end{aligned} \tag{8}$$

Hats denote the usual bias corrections. The decoupled decay in (8) is generally different from adding an ℓ_2 penalty to an adaptively preconditioned gradient [2]. Its diagonal adaptation omits cross-coordinate second moments, keeping the update inexpensive and separable across shards.

The matrix-aware methods can be written in the same outer form by making their state and transformation explicit:

$$S_t = \mathcal{A}(S_{t-1}, G_t), \qquad \widehat{D}_t = \mathcal{T}(S_t, Z_t), \qquad W_{t+1} = (1-\eta_t\lambda_t) W_t - \eta_t \widehat{D}_t, \tag{9}$$

Here S_t is statistical or numerical state, and Z_t may be a raw gradient, momentum, or corrected estimator. The map $\mathcal{T}$ may apply a preconditioner, transform singular values, or rescale its output. Even if it is linear for fixed S_t, the full gradient-to-update map can be nonlinear because S_t depends on the current gradient.

Operator and sign conventions. We call the operator acting directly on the vectorized gradient or momentum the update preconditioner. When a proof is written in a metric H_t, that preconditioner is H_t^{-1}; the moment matrices used to construct it are different objects. We reserve D_t for a direction subtracted in (9), and use Δ for a signed trial displacement in a linear-minimization problem. A complete algorithmic specification must also state the inputs to S_t and $\mathcal{T}$, averaging rule, damping placement, and refresh schedule. Substituting M_t for G_t in a Gram estimate, or a Nesterov combination for M_t in a polar map, changes the algorithm.

2.4 A factorized basis need not imply a factorized operator

For SPD factors, $D = L^{-a} Z R^{-b}$ gives $\operatorname{vec}(D) = (R^{-b} \otimes L^{-a}) \operatorname{vec}(Z)$. Coordinate scaling in a factorized orthogonal basis is more general:

$$P = Q \operatorname{diag}(\operatorname{vec}(s)) Q^\top, \qquad Q = Q_R \otimes Q_L, \tag{10}$$

The positive $m \times n$ array s is unrestricted in this expression. A single Kronecker operator in the same basis requires $s_{ij} = a_i b_j$, which fails, for example, for $\left(\begin{smallmatrix} 1 & 1 \\ 1 & 2 \end{smallmatrix}\right)$. SOAP can therefore retain a Kronecker-factored basis while learning adaptive scales that are not separable across the two modes [5, 10].

This suggests treating the coordinate system and the scaling rule as separate choices. For orthogonal U_t, V_t, write

$$\widetilde{Z}_t = U_t^\top Z_t V_t, \qquad D_t = U_t \mathcal{H}_t(\widetilde{Z}_t) V_t^\top. \tag{11}$$

Inverse powers produce separable scales in a factor eigenbasis, whereas SOAP uses entrywise adaptive scales. In the current input's singular-vector basis, polar and finite-polynomial maps specify a singular-value response. ARO combines adaptive rotation with normed steepest descent [12]; convergence results for rotation-based methods concern particular state and normalization recursions [13], stated in section 6.4 and appendix B.10. Table 1 summarizes these design choices.

Table 1: Design principles of matrix-aware optimization. Methods may combine several rows.

Principle	State retained	Transformation	Principal theoretical question
Coordinate adaptation	Entrywise moments	Diagonal scaling	Which coordinate geometry is favorable?
Mode adaptation	Mode Gram or fitted factors	One-/two-sided inverse powers	Which structured statistic and exponent are justified?
Basis adaptation	Mode bases and coordinate moments	Rotated diagonal scaling	How are scales and historical state transported?
Spectral update	Momentum or corrected gradient	Polar or finite polynomial	Which norm geometry and alignment are controlled?
Adaptive spectral update	Spectral input plus adaptive state	Pre/post scaling and orthogonalization	Does composition preserve a descent argument?
Compressed update	Subspaces, residuals, or low-rank state	Restricted spectral transformation	What information and convergence properties survive?

3 Structured adaptive preconditioning

3.1 From full-matrix adaptation to mode statistics

Full-matrix AdaGrad uses a cumulative outer-product matrix whose dimension equals the number of parameters [17]. Shampoo retains tensor-mode statistics instead. For a matrix block, its canonical cumulative update is

$$\begin{aligned} L_t &= \varepsilon I_m + \sum_{s=1}^{t} G_s G_s^\top, \\ R_t &= \varepsilon I_n + \sum_{s=1}^{t} G_s^\top G_s, \\ W_{t+1} &= W_t - \eta_t L_t^{-1/4} G_t R_t^{-1/4}. \end{aligned} \tag{12}$$

The corresponding vectorized metric is $H_t = R_t^{1/4} \otimes L_t^{1/4}$, giving the update $-\eta_t H_t^{-1} g_t$. For an order-k tensor, the original construction forms one factor per mode and uses exponent $-1/(2k)$ on each factor [3]. This preserves tensor structure without storing a full parameter-space second moment.

The fourth roots have a specific theoretical role. The original regret analysis compares the cumulative vectorized outer product with the product of mode contractions, leading to a bound involving gradient rank and traces of fourth roots. We develop that argument in section 6.2 and appendix A.1. A statistical whitening objective is a different starting point and need not select the same exponent.

One-sided updates, such as $L_t^{-1/2} G_t$, offer another choice for highly rectangular blocks. ASGO and unified structured-preconditioner analyses treat this geometry separately [18, 19]. Removing one side changes both the operator class and the exponent, so comparisons with two-sided Shampoo must also match the radius and smoothness assumptions of their bounds.

3.2 Kronecker approximation and whitening

To see why the exponent depends on the objective, suppose the population second moment is exactly separable: $C = B \otimes A$ with $A, B \succ 0$. Its mode contractions satisfy

$$L = \operatorname{tr}(B) A, \qquad R = \operatorname{tr}(A) B, \qquad s = \mathbb{E}\,\|G\|_F^2 = \operatorname{tr}(A)\operatorname{tr}(B). \tag{13}$$

It follows that

$$C = \frac{R \otimes L}{s}, \qquad \operatorname{unvec}\Big(C^{-1/2} \operatorname{vec}(G)\Big) = \sqrt{s}\, L^{-1/2} G R^{-1/2}. \tag{14}$$

Thus exact whitening by $C^{-1/2}$ uses inverse square roots, with a trace-dependent scale. Applying inverse fourth roots to the same Gram matrices does not give that whitening transform. The distinction separates the regret geometry of the original Shampoo construction from a statistical objective for its factors. Morwani et al. [20] develop the latter viewpoint by relating Shampoo's factors to a procedure for approximating the gradient second-moment matrix.

3.3 SOAP: separate the basis from the scales

Shampoo's factors determine both a basis and the scales applied in it. SOAP separates these choices. Write $L_t = Q_{L,t}\Lambda_{L,t}Q_{L,t}^\top$ and $R_t = Q_{R,t}\Lambda_{R,t}Q_{R,t}^\top$. During an interval with fixed bases, SOAP rotates the gradients and applies Adam-style adaptation in those coordinates [5]:

$$\begin{aligned}
\widetilde{G}_t &= Q_L^\top G_t Q_R, \\
\widetilde{M}_t &= \beta_1 \widetilde{M}_{t-1} + (1-\beta_1)\widetilde{G}_t, \\
\widetilde{V}_t &= \beta_2 \widetilde{V}_{t-1} + (1-\beta_2)\widetilde{G}_t^{\odot 2}, \\
D_t &= Q_L \left(\frac{\widehat{\widetilde{M}}_t}{\sqrt{\widehat{\widetilde{V}}_t} + \varepsilon} \right) Q_R^\top.
\end{aligned} \tag{15}$$

The basis can now be refreshed less often than the coordinatewise moments. Moreover, the dense array $\widetilde{V}_t$ allows the nonseparable scales in (10); SOAP is not restricted to correcting a separable pair of eigenvalue factors.

A basis refresh raises a further question: what happens to the accumulated moments? A full first moment can remain in the original coordinates and be projected into each new basis. In Algorithm 3 of the SOAP paper, the diagonal second-moment array continues its entrywise EMA across refreshes [5, Section 4.1]. Its entries consequently mix observations expressed in successive bases rather than reconstructing the historical second moment. Permuting, resetting, or approximately rotating this state defines a different recurrence.

The inability to reconstruct that moment does not rule out convergence. Proposition 7.4 shows what information diagonal storage loses, while Sun et al. [13] prove convergence for a specified moving-basis EMA, even with current-gradient-dependent rotations. Their normalizer is $\sqrt{V_t + \varepsilon}$ rather than $\sqrt{V_t} + \varepsilon$, and they retain the first moment in the original coordinates. The precise recurrence and schedules appear in section 6.4 and appendix B.10; they should not be replaced by the fixed-basis template above.

Eigenvalue correction and grafting. Scale changes also arise outside the basis update. Learning-rate grafting replaces D_t by

$$D_t^{\text{graft}} = \frac{\left\| D_t^{\text{base}} \right\|_F}{\|D_t\|_F} D_t \tag{16}$$

when the denominator is nonzero, with an implementation-specific rule at zero. It keeps a block's direction and imports another method's magnitude. Eschenhagen et al. [21] separate this effect from stale eigenbases and eigenvalue correction, making the update spectrum, rather than the basis alone, the object of comparison.

3.4 KL-based fitting and the statistical meaning of the factors

Instead of taking the mode contractions as the factors, one can specify a fitting objective. For a positive-definite second moment C and a separable approximation $B \otimes A$, the zero-mean Gaussian divergence $D_{\mathrm{KL}}(\mathcal{N}(0, C) \| \mathcal{N}(0, B \otimes A))$ is, up to constants and a factor $1/2$,

$$J(A, B) = n \log\det A + m \log\det B + \operatorname{tr}\left[(B^{-1} \otimes A^{-1}) C \right]. \tag{17}$$

Its block stationary conditions are

$$A = \frac{1}{n}\mathbb{E}[GB^{-1}G^\top], \qquad B = \frac{1}{m}\mathbb{E}[G^\top A^{-1}G]. \tag{18}$$

Each mode factor is estimated using the other mode's geometry. This coupling distinguishes the conditions from independent Gram accumulation; appendix A.2 gives the derivation. Since $(A, B) \mapsto (cA, B/c)$ leaves the Kronecker product unchanged, a normalization convention is needed to fix the factor scales.

KL-Shampoo and KL-SOAP use this objective to redesign factor estimation and basis/scale updates [10]. The objective also makes the modeling choice explicit: an uncentered second moment is being used as the covariance of an auxiliary zero-mean Gaussian. Other Bregman objectives lead to different fitting rules [22].

3.5 Practical variants

The practical variants address different costs of maintaining and applying the factors. Sketchy reduces the stored statistical information through Frequent Directions sketches of the mode statistics; its online-convex memory–regret bound depends on the discarded spectrum [23]. Root-free and inverse-free Shampoo take a different route, changing the factor recurrence to avoid explicit roots and inversions [24]. Simply omitting the root solver from the original recurrence would not reproduce these methods.

Other changes concern the update made from that state. SOAP and eigenvalue-corrected variants alter the represented operator. SPlus combines a historical eigenbasis with bounded instantaneous normalization, shape-aware scaling, and iterate averaging [25], changing the magnitude and averaging of the update as well as its basis.

Execution changes must be distinguished from both. Scalable and distributed Shampoo use blocking and delayed expensive operations [4, 26]: blocking can change the geometry, whereas delayed computation changes which numerical state is current. FOAM links damping to refresh frequency and operator error [27]. Reparametrized subspace updates and bfloat16 storage change the state's numerical representation [28], while DASH improves block-preconditioning and inverse-root execution [29].

Four-bit Shampoo illustrates why memory savings alone do not identify the approximation being made. It quantizes the eigenbasis and corrects the resulting loss of orthogonality in the inverse-root preconditioner [30]. Unlike Sketchy's rank truncation of the statistic, this is a numerical representation error; the four-bit designation does not apply to every state or arithmetic operation. The guarantees compared in section 6 attach to these particular recurrences, not simply to the name Shampoo.

4 Spectral and adaptive spectral updates

4.1 Idealized Muon and its norm geometry

An idealized Muon step applies the polar transformation to a momentum-like matrix Z_t:

$$D_t = a_t \,\mathrm{polar}(Z_t), \qquad W_{t+1} = (1 - \eta_t\lambda_t)W_t - \eta_t D_t. \tag{19}$$

The factor a_t includes a block or shape scale. Orthogonalization acts on the update input, not on W_t, so it does not generally constrain the weights to a Stiefel manifold. For a rank-deficient input, the canonical transformation is the partial isometry defined in (2).

The polar direction follows from linear minimization over a spectral-norm ball [6, 7]:

$$\min_{\|\Delta\|_{\mathrm{op}} \le \rho} \langle G, \Delta \rangle = -\rho \, \|G\|_* , \qquad \Delta^* = -\rho \, \mathrm{polar}(G). \tag{20}$$

Nuclear–spectral norm duality supplies the lower bound, and the compact SVD attains it. The same direction appears in a squared-norm formulation,

$$\arg\min_{\Delta} \left\{ \langle G, \Delta \rangle + \frac{1}{2\eta} \|\Delta\|_{\mathrm{op}}^2 \right\} \ni -\eta \, \|G\|_* \, \mathrm{polar}(G). \tag{21}$$

but with a different magnitude. Normalized descent, squared-norm descent, and preconditioned gradient descent therefore specify different updates even when their directions agree.

Momentum conventions. A two-coefficient template makes the input to polar explicit:

$$M_t = \beta_1 M_{t-1} + (1 - \beta_1) G_t, \qquad Z_t = \beta_2 M_t + (1 - \beta_2) G_t. \tag{22}$$

The choice $\beta_2 = 1$ gives a heavy-ball input; $\beta_2 = \beta_1$ gives a common normalized Nesterov-style combination. Some implementations instead keep the unnormalized buffer $B_t = \beta B_{t-1} + G_t$. With matching initialization and fixed coefficients, inputs may differ only by a positive scalar, which exact polar removes. Normalization floors, varying coefficients, or adaptive post-scaling can break that equivalence.

4.2 The actual finite-polynomial update

Muon avoids repeated SVD by using a short approximate orthogonalization procedure [6]. A representative fifth-degree iteration is

$$X_0 = Z/\tau, \qquad X_{j+1} = a_j X_j + b_j X_j X_j^\top X_j + c_j (X_j X_j^\top)^2 X_j. \tag{23}$$

The products can be ordered to use the smaller Gram matrix. In exact arithmetic, writing the input in its singular-vector basis gives

$$X_K = U \, \mathrm{diag}(q_K(\sigma_i/\tau)) \, V^\top, \qquad q_K = \phi_{K-1} \circ \cdots \circ \phi_0, \tag{24}$$

where $\phi_j(s) = a_j s + b_j s^3 + c_j s^5$. The prescribed finite composition is therefore a spectral filter. For the original empirical triple $(3.4445, -4.7750, 2.0315)$, often used for five steps [6, 31], a unit singular value is mapped to 0.701. One is not a fixed point, so this is not a classical iteration converging to one as the step count increases.

This observation leads to two different analyses. Taylor-family Newton–Schulz (NS) results control approximation to polar under spectral conditions [32]. Direct finite-map analyses instead bound the alignment and energy of the computed output [31, 33, 34]. Such a map can support descent despite a large worst-case polar error. Conversely, accurate polar computation does not remove momentum tracking error, stochastic noise, or the need for an appropriate learning-rate schedule.

Polar Express develops optimized matrix-sign/polar procedures and finite-precision analysis [35]. These numerical guarantees address the transform itself. An outer-optimizer guarantee must additionally connect its output to the gradient process.

4.3 Scale, parameter classes, and adaptation

The norm geometry fixes a direction, but not a scale suitable for every parameter block. For a full-rank $m \times n$ input with $r = \min(m, n)$,

$$\mathrm{RMS}(\mathrm{polar}(Z)) = \frac{\|\mathrm{polar}(Z)\|_F}{\sqrt{mn}} = \frac{1}{\sqrt{\max(m, n)}}. \tag{25}$$

A factor proportional to $\sqrt{\max(m, n)}$ consequently matches entrywise RMS across full-rank matrix shapes. Rank deficiency, finite polynomials, or a different target update distribution change the calculation. Scalable Muon revisits scaling and weight decay for large language models [8]; its learning rates must be interpreted together with the update scales.

Adaptation can act at several points in this construction. AdaGO adjusts only the output magnitude, using clipped current and accumulated gradient norms while preserving the polar direction [36]. Variance-Adaptive Muon's Muon-NSR and Muon-VS instead modulate momentum with variance information before orthogonalization [37]. Muon^2 applies Adam-style second-moment preconditioning at that stage, with Muon^2-F factorizing the additional state [38]. Input adaptation changes both the matrix seen by the spectral map and the statistical memory carried between steps.

One can also change the spectral response or rescale the output by coordinate. BeyondMuon studies $U\Sigma^p V^\top$ transformations of momentum or RMS-normalized inputs, varying how much singular-value contrast is retained [39]; it does not establish a universally best exponent. AdaMuon combines elementwise second-moment adaptation of orthogonalized directions with, in version 3, a sign transformation of momentum before orthogonalization [40]. NorMuon uses neuron-wise normalization/adaptation and a distributed implementation [41]. Unlike scalar output adaptation, coordinate-dependent scaling generally removes polar's equal-singular-value property, so the composed map becomes the object of analysis.

The order of these operations matters. Mousse orthogonalizes in coordinates determined by Kronecker-factored statistics [42]. The weighted spectral-ball construction in proposition 5.3 includes both the input coordinate change and the map back. Since $\mathrm{polar}(AZB)$ is generally not $A\,\mathrm{polar}(Z)B$, convergence has to be established for the chosen composition rather than inferred from its components independently.

Parameter roles add another design choice. A training recipe may assign Muon to selected matrices and another optimizer to embeddings, biases, or normalization parameters. Kimi K2's technical report describes a stabilization-oriented recipe at much larger scale [43], providing evidence about a complete training system. AF-Muon uses a support-aware finite-cap rule for tied vocabulary tables and RMS normalization for one-dimensional parameters [44]. Both examples make the assignment of geometry within a model part of the optimizer specification.

4.4 Allocating geometry across subspaces, layers, and time

A method need not apply one geometry to an entire parameter block. Dion uses low-rank orthonormalized updates with state and communication mechanisms for distributed training [45], and Dion3 extends the approach across the training stack [46]. COSMOS makes a different allocation: SOAP-like adaptation acts in a tracked leading subspace, while a Muon-style update acts on its complement [47]. These choices require attention to how omitted directions are handled—through residuals, tracked subspaces, or another geometry. Rank alone neither determines the update nor establishes equality to exact Muon; in COSMOS, the mixture and normalization also affect geometry and memory cost.

The transformed object can also change across layers. TEON groups compatible layer gradients into a higher-order tensor and chooses a matricization for orthogonalization [48]. This changes the geometry rather than merely batching independent matrix operations. LionMuon allocates spectral computation over time, alternating periodic spectral steps with sign steps while sharing momentum and using branch-specific step sizes [49]. The schedule changes the recurrence as well as the frequency of orthogonalization.

LiMuon combines recursive variance reduction with randomized-SVD compression of the carried estimator [50]. Here the stochastic oracle changes too: the guarantee uses paired, same-sample gradient evaluations, whose additional work must be counted in sample-complexity and runtime comparisons.

The theory has developed along these distinctions. Early work studies idealized Muon and its stochastic behavior [51]; subsequent analyses treat convergent NS families, empirical finite polynomials, Nesterov tracking, and variance-reduced estimators [31–34]. Deterministic regularized proxies admit Lyapunov analyses [52], while nonconvergence constructions and approximation-error analyses delimit particular settings [53]. Section 5 develops the algebraic links before section 6 compares the guarantees for each recurrence and oracle.

4.5 Comparing the core update rules

Table 2 collects the four core rules for an $m \times n$ block. It describes the gradient-driven direction in (9); learning rate, output scaling, and decoupled decay remain outer choices. The fuller state, numerical, and distributed specifications appear in appendix C.

Table 2: Core update rules after fixing the current optimizer state. A fixed-state linear form does not make the full gradient-to-update map linear when that state depends on the current gradient.

Method	Retained information	Representative transformation	Main distinction
AdamW	Entrywise first and second moments	Coordinatewise scaling of momentum, (8)	Scaling in the original coordinates
Shampoo	Historical mode Gram matrices	Two-sided inverse powers, (12)	A separable update operator
SOAP	Mode bases and coordinatewise moments	Rotate, adapt, and rotate back, (15)	Factored basis; generally nonseparable scales
Finite Muon	Momentum with a declared input convention	A fixed-depth singular-value map, (24)	Spectral transformation rather than covariance estimation

5 Algebraic connections and optimization dynamics

5.1 Instantaneous and accumulated adapted-polar identities

The simplest connection between Shampoo and Muon appears when the preconditioner is built from the very matrix to which it is applied.

Proposition 5.1 (Instantaneous identity and its damped filter)**.** *For any G, with inverse powers restricted to its singular support,*

$$(GG^\top)^{-1/4}G(G^\top G)^{-1/4} = \mathrm{polar}(G). \tag{26}$$

With the same damping $\delta > 0$ on both Gram factors, the singular response becomes

$$(\sigma^2+\delta)^{-1/4}\sigma(\sigma^2+\delta)^{-1/4} = \frac{\sigma}{\sqrt{\sigma^2+\delta}}. \tag{27}$$

Proof. Substitute the compact SVD and cancel the singular-value powers on the support. On the nullspace both sides of (26) are zero. The damped expression follows by applying the scalar function to each singular value. □

For matched inputs, the undamped instantaneous Shampoo direction is thus exactly polar [6, 7]. Historical accumulation and a separate momentum input remove the conditions behind this identity, while damping changes the response to (27). Accumulated statistics can still be related to polar through the following decomposition.

Proposition 5.2 (An adapted-polar decomposition)**.** *For SPD L, R, real exponents a, b, and any matrix M,*

$$L^{-a} M R^{-b} = \left[L^{-a}(MM^\top)^{1/4}\right] \operatorname{polar}(M) \left[(M^\top M)^{1/4} R^{-b}\right]. \tag{28}$$

Proof. The compact SVD gives $(MM^\top)^{1/4} \operatorname{polar}(M)(M^\top M)^{1/4} = M$. Multiplication by the two outside factors proves the claim without a commutativity assumption. □

The adapted-polar viewpoint of Eschenhagen et al. [9] accommodates historical statistics in this form. The bracketed factors need not be symmetric or positive definite. They combine the singular structure of the current M with the possibly nonstationary history in L, R, so the decomposition does not imply identical training trajectories.

There is also a short-sided instantaneous limit. For full-row-rank $G \in \mathbb{R}^{m\times n}$ with $m \le n$,

$$\left(\frac{1}{n} G G^\top\right)^{-1/2} G = \sqrt{n} \operatorname{polar}(G). \tag{29}$$

The short-sided KL connection in version 10 of Lin et al. [10] uses this limit. It should be read separately from the population fitting problem in (17) and the online solver used to estimate its factors.

5.2 Weighted spectral geometry and composition order

A coordinate change provides a second connection: polar can define steepest descent in a weighted spectral geometry, provided the output is transformed back as well.

Proposition 5.3 (Spectral descent in anisotropic coordinates)**.** *Let $A \succ 0$, $B \succ 0$, and $\rho > 0$. A minimizer of*

$$\min_{\Delta} \langle G, \Delta \rangle \quad \textit{subject to} \quad \left\| A^{1/2} \Delta B^{1/2} \right\|_{\mathrm{op}} \le \rho \tag{30}$$

is

$$\Delta^* = -\rho A^{-1/2} \operatorname{polar}(A^{-1/2} G B^{-1/2}) B^{-1/2}. \tag{31}$$

Proof. Set $Y = A^{1/2}\Delta B^{1/2}$. The objective is $\left\langle A^{-1/2} G B^{-1/2}, Y \right\rangle$, so (20) applies, followed by the inverse coordinate transformation. □

The two outside factors in (31) are part of the solution, not optional post-processing. Omitting them gives a different method. This construction interprets curvature-aware spectral compositions such as Mousse [42], without identifying gradient-estimated A, B with curvature factors.

Equivariance. For orthogonal U, V, $\operatorname{polar}(UZV^\top) = U \operatorname{polar}(Z) V^\top$. Exact Shampoo has the same left/right orthogonal equivariance when initialization is isotropic and the history is transformed consistently. This symmetry does not extend automatically to arbitrary rotations of the flattened mn-dimensional space, which can destroy matrix/Kronecker structure. Coordinatewise scaling generally has less invariance. For basis-dependent methods, repeated eigenspaces also make eigenbasis choices nonunique; an operator-level statement must account for that freedom. Rotation claims therefore need both the transformation group and the rule for transforming the state.

5.3 A frozen quadratic separates curvature from noise

Algebraic identities compare updates at a fixed state. To examine their effect over time, consider a stochastic quadratic with a fixed preconditioner:

$$f(x) = \tfrac{1}{2}x^\top Hx, \qquad x_{t+1} = x_t - \eta P(Hx_t + \xi_t), \tag{32}$$

Here $H, P \succ 0$ are fixed. Let x_t be measurable with respect to a filtration $\mathcal{H}_t$, let ξ_t be $\mathcal{H}_{t+1}$-measurable, and assume $\mathbb{E}[\xi_t \mid \mathcal{H}_t] = 0$ and $\mathbb{E}[\xi_t\xi_t^\top \mid \mathcal{H}_t] = \Sigma$ for a fixed finite matrix $\Sigma \succeq 0$. The initial state is deterministic or satisfies $\mathbb{E}\,\|x_0\|^2 < \infty$. These assumptions fix the conditional noise covariance without requiring temporal independence. Define

$$z = P^{-1/2}x, \qquad \widetilde{H} = P^{1/2}HP^{1/2}, \qquad \widetilde{\Sigma} = P^{1/2}\Sigma P^{1/2}. \tag{33}$$

Then $z_{t+1} = (I - \eta\widetilde{H})z_t - \eta P^{1/2}\xi_t$.

Proposition 5.4 (Stability, finite-time loss, and stationary loss)**.** *The deterministic recursion is strictly stable precisely when $0 < \eta < 2/\lambda_{\max}(\widetilde{H})$. In that range, its noise-driven second moment converges to the solution of*

$$S = (I - \eta\widetilde{H})S(I - \eta\widetilde{H}) + \eta^2\widetilde{\Sigma}. \tag{34}$$

Let $Q^\top\widetilde{H}Q = \operatorname{diag}(h_i)$, $\nu_i = (Q^\top\widetilde{\Sigma}Q)_{ii}$, and $b_i = \mathbb{E}[(Q^\top P^{-1/2}x_0)_i^2]$. For every integer $t \geq 0$ in the stable stepsize range,

$$\mathbb{E}f(x_t) = \frac{1}{2}\sum_i h_i(1 - \eta h_i)^{2t}b_i + \frac{\eta}{2}\sum_i \frac{\nu_i\big[1 - (1 - \eta h_i)^{2t}\big]}{2 - \eta h_i}. \tag{35}$$

The first term is the initial-state contribution; for deterministic x_0 it is the loss of the mean trajectory. The second term is the accumulated-noise contribution. Consequently,

$$\lim_{t\to\infty} \mathbb{E}f(x_t) = \frac{\eta}{2}\sum_i \frac{\nu_i}{2 - \eta h_i}. \tag{36}$$

Proof. Each deterministic eigenmode has multiplier $a_i = 1 - \eta h_i$. Strict stability is equivalent to $|a_i| < 1$. Conditional centering eliminates state–noise cross terms, so, for $y_t = Q^\top z_t$, $\mathbb{E}[y_{t,i}^2] = a_i^{2t}b_i + \eta^2\nu_i\sum_{k=0}^{t-1} a_i^{2k}$. The geometric sum and $1 - a_i^2 = \eta h_i(2 - \eta h_i)$ give (35) after multiplication by $h_i/2$ and summation. The full second-moment recursion likewise has a convergent matrix series with unique limit (34). Since each b_i is finite, taking $t \to \infty$ proves (36). □

Equation (35) exposes two competing effects of preconditioning: contraction of the initial error and accumulation of noise. The covariance of the update noise is $\eta^2 P\Sigma P$ in the original coordinates and $\eta^2\widetilde{\Sigma}$ in the z coordinates. Statements about noise amplification must specify which covariance is measured. Even with that choice fixed, separate curvature and noise spectra do not determine the loss.

Example 5.5 (Identical spectra, different stationary losses)**.** Take $P = I$, $H = \operatorname{diag}(1, 3)$, and $\eta = 1/2$. For $\Sigma_1 = \operatorname{diag}(1, 0)$, (36) is $1/6$; for $\Sigma_2 = \operatorname{diag}(0, 1)$, it is $1/2$. The covariance spectra are identical, as is the curvature spectrum, but their relative orientation changes the loss floor by a factor of three.

The difference comes entirely from relative orientation. The model thus motivates studying curvature and noise together rather than optimizing either spectrum in isolation. Extending it to adaptive P_t would introduce changing coordinates and, with current-sample dependence, noise correlations. Those effects and Muon's nonlinear input map are outside this calculation, as are claims about the generalization effects of reducing blockwise spectral disparity.

6 Comparing theoretical guarantees

6.1 Assumptions and convergence measures

A convergence rate becomes comparable only after its algorithm, objective class, geometry, oracle, schedule, and controlled quantity have been specified. We write $\Delta = F(W_1) - F_* > 0$, use L_F for Frobenius smoothness, and reserve L_{op} for spectral-to-nuclear smoothness:

$$\|\nabla F(X) - \nabla F(Y)\|_* \le L_{\mathrm{op}} \|X - Y\|_{\mathrm{op}} \tag{37}$$

This is a different assumption from Frobenius Lipschitz continuity. Converting between the norms can introduce dimension or rank factors; adaptive and non-Euclidean geometries require further distinctions [11].

The conclusions also differ. For example,

$$\frac{1}{T}\sum_t \mathbb{E}\,\|\nabla F(W_t)\|_F^2 \le \varepsilon^2, \quad \frac{1}{T}\sum_t \mathbb{E}\,\|\nabla F(W_t)\|_* \le \varepsilon, \quad \|\nabla F(W_t)\|_F \to 0 \text{ a.s.} \tag{38}$$

are three separate statements. An average bound applies to a uniformly sampled iterate, not necessarily to the last iterate or to convergence of the parameters. Convex objective-gap and online regret bounds have their own interpretations. Table 3 collects the main results, with the assumptions and schedules recorded in appendix B.

Table 3: Representative proof map. The detailed assumptions and schedules in appendix B are essential to each row; rates are not a cross-method ranking.

Anchor	Controlled quantity	Main proof ingredient	Main coverage boundary
Original Shampoo [3]	Convex regret	Monotone metric potential and rank-sensitive domination	Cumulative statistics; exact powers
Structured / ASGO [18, 19]	Regret or convex gap	Restricted metric families; mode smoothness and noise traces	Selected one-sided/structured geometry; radius conditions
Two geometries [11]	Induced dual-gradient norm	Adaptive smoothness; structured trace bounds	Strong almost-sure noise envelope; selected metric cones
AdamW-style Shampoo v4 [54]	Nuclear gradient; generalized power measure	Matrix inequalities; momentum tracking	Coupled decay regime; fixed EMA for no-decay extension

Table 3 continued

Anchor	Controlled quantity	Main proof ingredient	Main coverage boundary
Generalized SOAP [13]	Average nuclear gradient	Row–column trace control	Exact stated recursion, damping and schedule; no transport-quality guarantee
Taylor NS Muon [32]	Average nuclear gradient	Polar residual and momentum tracking	Particular Taylor family; spectral residual controls
Practical Muon [31]	Average Frobenius gradient	Actual polynomial and input tracking	Five empirical steps; horizon-dependent floor, momentum, decay
Layer-wise finite Muon [33]	Sum of average block Frobenius gradients	Joint descent, scalar-map alignment/energy, Nesterov tracking	Coupled schedules; exact arithmetic; constants fixed for dimension claim
MVR1 / MVR2 [34]	Average Frobenius gradient; PL gap with extras	Recursive estimator; finite-map descent	Changed oracle for MVR2; stronger assumptions for PL
Regularized proxy [52]	Deterministic gradient limit; PL linear gap	Bounded preconditioner and Lyapunov function	Soft proxy and admissible parameters, not equality to finite NS

6.2 The original Shampoo regret mechanism

Let f_t be convex losses, $G_t \in \partial f_t(W_t)$, and $\mathcal{R}_T = \sum_{t=1}^{T}[f_t(W_t) - f_t(W_*)]$. Suppose $r \geq 1$, $\operatorname{rank}(G_t) \leq r$, and $\|W_t - W_*\|_F \leq D$. For the cumulative update (12), the original analysis gives the following bound [3, Theorem 7]:

Theorem 6.1 (Cumulative Shampoo regret, common notation)**.** *With exact matrix powers, fixed stepsize $\eta > 0$, and the assumptions above,*

$$\mathcal{R}_T \leq \left(\frac{D^2}{2\eta} + \eta r\right) \operatorname{tr}(L_T^{1/4}) \operatorname{tr}(R_T^{1/4}). \tag{39}$$

Choosing $\eta = D/\sqrt{2r}$ gives

$$\mathcal{R}_T \leq \sqrt{2r}\, D\, \operatorname{tr}(L_T^{1/4}) \operatorname{tr}(R_T^{1/4}). \tag{40}$$

Metric projection onto a bounded convex domain can enforce the distance condition. Without projection, it must hold along the iterates; either way, the optimized stepsize in (40) requires a radius bound.

The proof shows where the mode structure enters. Set $g_t = \operatorname{vec}(G_t)$, $H_t = R_t^{1/4} \otimes L_t^{1/4}$, and $S_t = \varepsilon I + (1/r)\sum_{s\leq t} g_s g_s^\top$. Rank-sensitive domination gives

$$S_t \preceq I_n \otimes L_t, \qquad S_t \preceq R_t \otimes I_m, \qquad S_t^{1/2} \preceq H_t. \tag{41}$$

The last inequality uses monotonicity of the positive matrix geometric mean and of the square root. A variable-metric potential bounds the displacement contribution by $D^2 \operatorname{tr}(H_T)/(2\eta)$, while trace

concavity gives $\sum_t g_t^\top H_t^{-1} g_t \le 2r \operatorname{tr}(H_T)$. Combining the two proves (39); appendix A.1 supplies the intermediate steps.

Cumulative factors increase in Loewner order, which allows the potential to telescope. EMA factors can shrink, leaving a positive-metric-variation term instead. This is why the cumulative proof does not transfer unchanged to EMA. Grafting, momentum, stale powers, and rounding introduce further quantities to control.

6.3 Structure-sensitive convex results and practical nonconvex Shampoo

One-sided structured adaptation. Xie et al. [19] analyze algebraically structured adaptive-preconditioner families. Their one-sided regret bound is proportional to

$$D_{\mathrm{op}} \operatorname{tr}\left(\left[\sum_{t=1}^{T} G_t G_t^\top\right]^{1/2}\right), \tag{42}$$

plus initialization terms, under explicit factor normalization and domain-radius conditions. The matrix-subalgebra framework unifies its admissible families; a two-sided separable operator still has to satisfy the relevant membership conditions.

ASGO makes the selected mode's curvature and noise dependence explicit [18, Theorem 3]. Suppose F is convex and one-smooth in $\|X\|_A^2 = \operatorname{tr}(X^\top A X)$ for SPD A, and the single-sample noise obeys $\mathbb{E}[NN^\top] \preceq V^2$. With minibatch size B, the prescribed stepsize and damping, and bounded iterate radii D_{op}, D_F, the bound is

$$\frac{1}{T}\sum_{t=0}^{T-1} \mathbb{E}[F(W_t) - F(W_*)] \le \frac{4D_{\mathrm{op}}^2 \operatorname{tr}(A)}{T} + \frac{2\sqrt{2} D_{\mathrm{op}} \operatorname{tr}(V)}{\sqrt{BT}} + \frac{2\varepsilon D_F^2}{D_{\mathrm{op}} T}. \tag{43}$$

Curvature and noise enter through mode-wise traces. Low effective spectral complexity can improve these quantities relative to isotropic bounds, but the improvement must be assessed with the same geometry and radius assumptions.

Adaptive and non-Euclidean smoothness. Xie et al. [11, Theorems C.7–C.8] obtain $\widetilde{\mathcal{O}}(T^{-1/4})$ stationarity bounds for specified adaptive families, including one-sided Shampoo, in nonconvex cumulative and EMA settings. Their adaptive smoothness optimizes over admissible quadratic geometries, whereas normalized steepest descent uses a fixed norm. The noise assumptions are also consequential: Assumption C.1 requires an almost-sure two-sided envelope on stochastic gradient outer products, stronger than a conditional covariance bound. Theorems 4.6 and 4.9 distinguish adaptive-noise upper bounds from dimension-dependent standard-noise lower bounds for sign descent. Applying these comparisons to two-sided factors or finite-NS maps requires checking both the family and the oracle assumptions.

EMA, momentum, and decoupled decay. Li et al. [54] address a practical-form Shampoo recurrence directly:

$$\begin{aligned} M_t &= \theta M_{t-1} + (1-\theta)G_t, \\ L_t &= \beta L_{t-1} + (1-\beta)G_t G_t^\top, \\ R_t &= \beta R_{t-1} + (1-\beta)G_t^\top G_t, \\ W_{t+1} &= (1-\eta\lambda)W_t - \eta(L_t+\varepsilon I)^{-1/(2p)} M_t (R_t+\varepsilon I)^{-1/(2q)}. \end{aligned} \tag{44}$$

For $p, q \in [1,\infty]$ with $p^{-1}+q^{-1}=1$, the symmetric choice $p=q=2$ gives fourth roots and an endpoint gives a one-sided update. We use $1/\infty = 0$ and interpret a zero exponent as the identity. Under Frobenius smoothness, conditional unbiasedness, matrix noise envelopes, a lower eigenvalue bound $\widehat{\varepsilon}$, and the coupled parameters in appendix B.4, Theorem 1 of version 4 gives

$$\frac{1}{T}\sum_{t=1}^{T} \mathbb{E}\left\|\nabla F(W_t)\right\|_* \le \left(8\sqrt{m+n} + \frac{119\widehat{\sigma}}{\sqrt{\widehat{\varepsilon}}}\right) \max\left\{ \left(\frac{\sigma^2 L_F \Delta}{T}\right)^{1/4}, \sqrt{\frac{L_F\Delta}{T\gamma}} \right\}. \tag{45}$$

Here $\gamma \in (0,1)$ and $\widehat{\sigma}^2 = \max\{\sigma^2, L_F\Delta/(T\gamma^2)\}$. The learning rate, momentum, damping, decay, and initialization are jointly constrained.

Theorem 2 treats a different regime: $p^{-1}+q^{-1} = 2-\omega$, $0 < \omega \le 2$, with any fixed second-moment EMA coefficient $\beta \in (0,1)$. It sets $\lambda = 0$ and retains horizon-dependent first-moment momentum, stepsize, and damping. The controlled quantity is $\sum_i \sigma_i(\nabla F)^\omega$, with order $\mathcal{O}(T^{-\omega/4} + T^{-1/2})$ when problem constants are fixed. Together, these results cover specific extensions beyond cumulative Shampoo, while leaving arbitrary fixed schedules with decay, cached powers, and mixed precision to separate analyses.

6.4 Rotation-based convergence and information transport

Sun et al. [13, Algorithm 4; Theorem 1] analyze the following generalized SOAP recurrence:

$$\begin{aligned} M_t &= \theta M_{t-1} + (1-\theta)G_t, \\ V_t &= \beta V_{t-1} + (1-\beta)(Q_{L,t}^\top G_t Q_{R,t})^{\odot 2}, \\ W_{t+1} &= W_t - \eta Q_{L,t} \frac{Q_{L,t}^\top M_t Q_{R,t}}{\sqrt{V_t+\varepsilon}} Q_{R,t}^\top. \end{aligned} \tag{46}$$

The first moment stays in the original coordinates, and the orthogonal rotations may depend on G_t. Under Frobenius smoothness, unbiased variance-bounded gradients, and the parameter schedule in appendix B.10,

$$\frac{1}{T}\sum_{t} \mathbb{E}\left\|\nabla F(W_t)\right\|_* \le \left(2\sqrt{m+n} + \frac{16\widehat{\sigma}}{\delta}\right)\left(\frac{\widehat{\sigma}^2 L_F \Delta}{T}\right)^{1/4}. \tag{47}$$

Row–column trace control establishes convergence without reconstructing the historical second moment. This resolves one concern about changing bases: exact recovery of that matrix is not required for this result. It leaves a different question open, because the bound also holds for identity rotations and therefore does not establish an advantage for learning the basis.

The theorem retains coordinate-indexed V_t across rotations and uses the displayed normalizer, without weight decay or bias correction. Transporting the stored state changes the recurrence.

The information-recovery result in proposition 7.4 concerns preservation of a historical operator, whereas this theorem concerns convergence of the stated update.

6.5 Exact polar, convergent NS, and actual finite maps

An exact-gradient baseline. For deterministic $D_t = \mathrm{polar}(\nabla F(W_t))$, Frobenius smoothness gives

$$F(W_{t+1}) \le F(W_t) - \eta \, \|\nabla F(W_t)\|_* + \tfrac{1}{2} L_F \eta^2 r. \tag{48}$$

Choosing a constant horizon-dependent stepsize yields an average nuclear-gradient bound $\sqrt{2L_F r \Delta / T}$. Under (37), the quadratic remainder becomes $L_{\mathrm{op}}\eta^2/2$, giving $\sqrt{2L_{\mathrm{op}}\Delta/T}$. The latter has no explicit rank factor, but uses a different smoothness constant. Stochastic momentum requires an additional input-tracking bound in either geometry.

Convergent Taylor-family NS. Kim and Oh [32, Theorems 1–2] analyze a specified Taylor polynomial family with operator-to-nuclear smoothness, unbiased minibatch gradients, and horizon-dependent momentum. Their stationarity estimate takes the form

$$\frac{1}{T}\sum_t \mathbb{E}\,\|\nabla F(W_t)\|_* \le \chi_K \, \mathcal{O}\left(\sqrt{\frac{L_{\mathrm{op}}\Delta}{T}} + \frac{\sigma r}{\sqrt{BT}} + \left(\frac{r\sigma^2 L_{\mathrm{op}}\Delta}{BT} \right)^{1/4} \right). \tag{49}$$

The factor χ_K depends on the initialized support residual and is uniformly bounded only if that residual stays uniformly below one. Appendix H.4 of the source excludes the empirical coefficient triple from the Taylor family. Equation (49) therefore concerns that convergent family, not the empirical quintic used in the next results.

Actual empirical polynomials. Wang et al. [31, Theorem 1] retain the five-step empirical polynomial, a Nesterov-style input, normalization floor, and decay. They obtain a nonconvex average Frobenius-gradient rate $\mathcal{O}(T^{-1/4})$ with schedules including $\eta \asymp T^{-3/4}$, $1-\beta \asymp T^{-1/2}$, a floor of order at most $T^{-1/4}$, and a prescribed coupled decay choice.

Peng et al. [33, Theorem 3.4 and Corollary 3.5] analyze layer-wise finite maps and Nesterov tracking under generalized smoothness, in exact arithmetic and without decay. Their coupled horizon-dependent coefficients yield a $T^{-1/4}$ average Frobenius bound. No matrix dimension or rank appears explicitly when the block count, scalar-map constants, smoothness, and noise are fixed, although those quantities may vary with network width.

Both approaches analyze the finite map's alignment and energy rather than requiring uniform approximation to polar. The following lemma captures that mechanism.

Lemma 6.2 (Finite-map alignment and descent)**.** *Let q be a scalar function with $q(0) = 0$ and $0 < \ell \le q(s)/s \le u$ for $0 < s \le 1$. For $Z \neq 0$, define $\mathcal{T}(Z) = U \operatorname{diag}(q(\sigma_i/\|Z\|_F))V^\top$ and $\mathcal{T}(0) = 0$. For any matrix g of the same shape,*

$$\begin{aligned} &\|\mathcal{T}(Z)\|_F \le u, \qquad \langle Z, \mathcal{T}(Z)\rangle \ge \ell \|Z\|_F, \\ &\langle g, \mathcal{T}(Z)\rangle \ge \ell \|g\|_F - (\ell + u)\|Z - g\|_F. \end{aligned} \tag{50}$$

If $g = \nabla F(W)$ and F is L_F-smooth, then

$$F(W - \eta\mathcal{T}(Z)) \le F(W) - \eta\ell\|g\|_F + \eta(\ell + u)\|Z - g\|_F + \tfrac{1}{2}L_F\eta^2u^2. \tag{51}$$

Proof. For $s_i = \sigma_i/\|Z\|_F$, $\sum_i s_i^2 = 1$. Thus $\sum_i q(s_i)^2 \le u^2$ and $\sum_i \sigma_i q(s_i) \ge \ell\|Z\|_F$. Subtract $\langle Z - g, \mathcal{T}(Z)\rangle$ and use $\|Z\|_F \ge \|g\|_F - \|Z - g\|_F$. The zero-input case obeys the same inequality. Smoothness proves the final line. □

For exact gradients, averaging (51) and optimizing the constant stepsize yields $(u/\ell)\sqrt{2L_F\Delta/T}$. Momentum introduces the tracking error $\|Z - g\|_F$. Scalar-map bounds for practical polynomials are available [33, 34], while a normalization floor needs a separate argument near zero.

6.6 Tracking, variance reduction, and asymptotic claims

Momentum trades fresh noise against lag in following a changing gradient. For $e_t = M_t - \nabla F(W_t)$, the EMA recurrence gives

$$e_t = \beta e_{t-1} + (1 - \beta)N_t + \beta(\nabla F(W_{t-1}) - \nabla F(W_t)). \tag{52}$$

A large β suppresses the fresh-noise term but retains drift. With bounded updates, typical noise and drift contributions scale as $\sqrt{1 - \beta}$ and $\eta/(1 - \beta)$, respectively. Controlling both explains the coupled schedules above; a theorem that sends β toward one does not also justify an arbitrary fixed value as T grows.

Variance reduction changes this balance by changing the estimator. Chang et al. [34, Theorems 6 and 8] obtain anytime rates $\widetilde{\mathcal{O}}(T^{-1/4})$ for MVR1 and $\widetilde{\mathcal{O}}(T^{-1/3})$ for MVR2, with practical finite NS and controlled decay. MVR2 requires mean-square smoothness of same-sample gradient differences and evaluates each fresh sample at two parameter locations. Its faster rate therefore includes additional oracle work. The PL-based last-iterate objective bounds further require bounded gradients along the iterates.

The Frobenius Polyak–Łojasiewicz (PL) condition, $\frac{1}{2}\|\nabla F(W)\|_F^2 \ge \mu_{\mathrm{PL}}(F(W) - F_*)$ for some $\mu_{\mathrm{PL}} > 0$ on the relevant domain, connects stationarity to objective error. It is an additional assumption. The soft polar proxy $(MM^\top + \varepsilon^2 I)^{-1/2}M$ admits a bounded-preconditioner heavy-ball interpretation and deterministic Lyapunov and PL arguments under specified parameter conditions [52]. This is a different map from the empirical finite polynomial.

Finally, nonconvergence constructions delimit particular Muon settings [53]. Even the scalar recurrence $x_{t+1} = x_t - \eta\operatorname{sign}(x_t)$ on $f(x) = x^2/2$ cycles between x_0 and $x_0 - \eta$ when $0 < x_0 < \eta$. Fixed normalized steps need not produce vanishing gradients, regardless of how accurately their direction is computed.

7 Inexact matrix computation

7.1 Matrix targets and numerical routines

A numerical routine does not identify an optimizer until its target is specified. An eigendecomposition $A = Q\Lambda Q^\top$ gives the SPD matrix function $A^{-p} = Q\Lambda^{-p}Q^\top$; an SVD gives a rectangular matrix's canonical polar factor. QR produces an orthogonal basis and triangular factor, and can track eigenvectors through orthogonal iteration. Newton–Schulz names a family of polynomial or rational recurrences, not a single target operator.

Proposition 7.1 (A QR basis is not generally a polar factor)**.** *If $Z = QR$ is a thin QR factorization with $Q^\top Q = I$ and R invertible, then* $\operatorname{polar}(Z) = Q \operatorname{polar}(R)$*. In particular, for*

$$Z = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}, \qquad Q = I, \qquad \operatorname{polar}(Z) = \frac{1}{\sqrt{5}} \begin{pmatrix} 2 & 1 \\ -1 & 2 \end{pmatrix} \neq Q. \tag{53}$$

Proof. Since $Z^\top Z = R^\top R$, $Z(Z^\top Z)^{-1/2} = QR(R^\top R)^{-1/2}$. For the displayed matrix, the proposed polar factor is orthogonal and its transpose times Z is SPD, which identifies the polar decomposition. □

SOAP- and KL-based implementations use QR to update a historical basis [5, 10]. A step $Q^+ = \operatorname{qr}(AQ)$ approximates an eigenbasis of A, with accuracy depending on the previous basis and spectrum. Caching that basis while updating the scales is consequently different from caching an inverse-root operator. Table 4 separates the numerical targets and the associated errors.

7.2 Fixed-depth polar approximation

The distinction between a finite spectral map and exact polar becomes unavoidable near vanishing singular values.

Table 4: Numerical targets and the approximation that must be controlled.

Target	Representative realization	Main error object	Dense cost scale
SPD inverse power	EVD or inverse-root solver	$\left\|\widehat{A^{-p}} - A^{-p}\right\|$	$O(d^3)$ for dense EVD
Moving eigenbasis	EVD refresh; warm-started QR iteration	Operator error and basis/subspace mismatch	$O(d^3)$ for a full QR step; less for restricted subspaces
Rectangular polar	SVD; convergent NS; Polar Express	Polar residual or singular response	$O(mn\min(m,n))$ per dense SVD-scale operation or polynomial step
Practical finite map	Prescribed polynomial with finite K	Alignment, energy, rounding of that map	$O(Kmn\min(m,n))$

Proposition 7.2 (A boundary of fixed-polynomial approximation)**.** *Let q be any fixed continuous scalar map with $q(0) = 0$. Normalize $Z_\varepsilon = \operatorname{diag}(1, \varepsilon)$ by its Frobenius norm and apply q to its singular values. Then*

$$\liminf_{\varepsilon \downarrow 0} \|\mathcal{T}(Z_\varepsilon) - \operatorname{polar}(Z_\varepsilon)\|_{\mathrm{op}} \geq 1. \tag{54}$$

Proof. The second normalized singular value is $\varepsilon/\sqrt{1+\varepsilon^2} \to 0$, so its transformed value tends to zero. For every $\varepsilon > 0$, the corresponding exact polar singular value is one. The spectral error is at least their absolute difference. □

Every fixed finite polynomial composition has this limitation. Increasing depth can still give convergence at a fixed input, but a fixed depth cannot handle arbitrarily small positive singular values uniformly. A uniform result therefore needs spectral bounds, adaptive work, or a different error measure.

Direct analyses of empirical Muon use the last option: scalar-map inequalities give descent through lemma 6.2 [31, 33, 34]. Rank-independent bounds may nevertheless be conservative, because the constants controlling $q(s)/s$ can grow with depth. If $|q(s)| \leq c$ as well, the update norm is bounded by $\min\{u, c\sqrt{r}\}$; the better estimate depends on both the depth and matrix shape.

For example, take $q_5 = \phi^{\circ 5}$ with the empirical quintic $\phi(s) = 3.4445s - 4.7750s^3 + 2.0315s^5$. Direct evaluation and the chain rule give

$$q_5'(0) = 3.4445^5 \approx 484.8763, \qquad q_5(1) \approx 0.6964364, \qquad \frac{u}{\ell} \geq \frac{q_5'(0)}{q_5(1)} \approx 696.225. \tag{55}$$

Any constants satisfying lemma 6.2 must obey $u \geq \lim_{s \downarrow 0} q_5(s)/s = q_5'(0)$ and $\ell \leq q_5(1)$. The ratio in (55) is thus a lower bound on this lemma's uniform constants, not on observed optimization error. Figure 1 shows the singular responses at several depths.

Turbo-Muon changes the initialization of Newton–Schulz to reduce inner work [55]. Such a change must be evaluated together with the initialized matrix, the resulting finite map, and outer-loop progress. Fewer inner iterations can change the update and need not reduce total training time.

7.3 Damping and spectral sensitivity

For inverse powers, damping gives an operator-level perturbation bound that does not require separated eigenvalues.

Proposition 7.3 (Damped inverse-power perturbation)**.** *For $A, B \succeq 0$, $\delta > 0$, and $0 < p \leq 1$,*

$$\left\|(A + \delta I)^{-p} - (B + \delta I)^{-p}\right\|_{\mathrm{op}} \leq p\delta^{-p-1} \|A - B\|_{\mathrm{op}}. \tag{56}$$

The resolvent proof in appendix A.3 handles noncommuting A, B; for $p = 1$, the inverse identity suffices. Clustered eigenvalues can allow large rotations of individual eigenvectors even when a smooth function of the full matrix remains stable. Small damping, on the other hand, amplifies operator sensitivity despite an accurate eigendecomposition. EVD error, eigenbasis tracking error, and inverse-root update error therefore measure different effects.

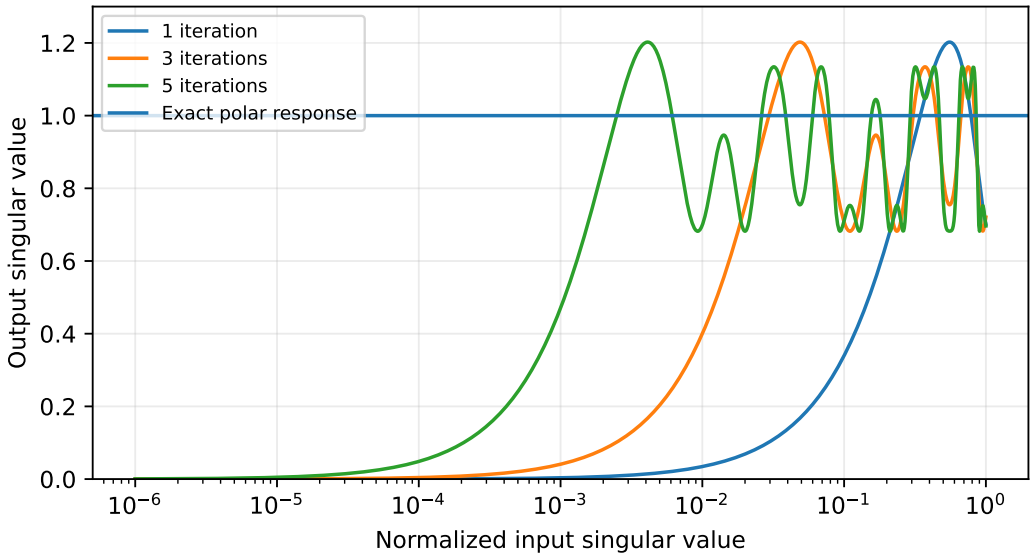


Figure 1: Analytic singular responses of the empirical Muon quintic after one, three, and five steps. One is not a fixed point of this map, and tiny input singular values remain suppressed at fixed depth.

For $D = AZB$ and $\widehat{D} = \widehat{A}Z\widehat{B}$,

$$\left\|\widehat{D} - D\right\|_F \leq \left\|\widehat{A} - A\right\|_{\mathrm{op}} \|Z\|_F \|B\|_{\mathrm{op}} + \left\|\widehat{A}\right\|_{\mathrm{op}} \|Z\|_F \left\|\widehat{B} - B\right\|_{\mathrm{op}}. \tag{57}$$

Combining (56) and (57) turns Gram-estimation errors into update errors, with constants that depend on damping. Purifying Shampoo, FOAM, low-precision reparametrization, and DASH address estimation, stale state, or numerical execution at different points in this calculation [21, 27–29]. As figure 2 illustrates, damping also changes the target operator near small eigenvalues.

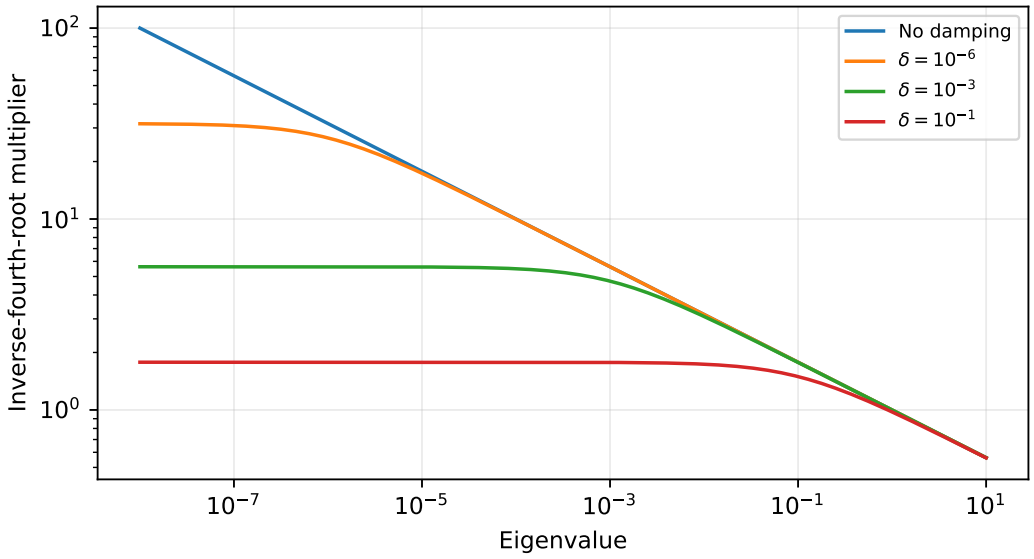


Figure 2: Analytic inverse-fourth-root multipliers at different damping values. Damping changes the target operator near small eigenvalues.

7.4 Information loss under basis changes

A basis refresh is not only a numerical approximation problem. When the optimizer stores a diagonal second moment, some information needed in the new basis has already been discarded.

Proposition 7.4 (Diagonal state cannot determine arbitrary rotated diagonals)**.** *There is no rule that, given only* $\mathrm{diag}(C)$ *and an arbitrary orthogonal* Q*, recovers* $\mathrm{diag}(Q^\top CQ)$ *for every positive-definite* C*.*

Proof. For $|\rho| < 1$, let

$$C_\rho = \begin{pmatrix} 1 & \rho \\ \rho & 1 \end{pmatrix}, \qquad Q = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}. \tag{58}$$

All inputs have diagonal $(1, 1)$, but $\mathrm{diag}(Q^\top C_\rho Q) = (1 + \rho, 1 - \rho)$. The proposed rule receives identical data for distinct required outputs. □

Corollary 7.5 (Exact diagonal transport for all moments)**.** *For a fixed orthogonal* $Q \in \mathbb{R}^{d\times d}$*, the vector* $\mathrm{diag}(Q^\top CQ)$ *is determined by* $\mathrm{diag}(C)$ *for every SPD* C *if and only if* Q *is a signed permutation matrix.*

Proof. Signed permutations only permute diagonal entries. Conversely, for distinct indices a, b, the SPD matrices $I \pm \epsilon(e_a e_b^\top + e_b e_a^\top)$, $0 < \epsilon < 1$, have the same diagonal. Equality of their required outputs forces $Q_{aj}Q_{bj} = 0$ for every column j. Thus each column has at most one nonzero entry. Unit column norms and orthogonality force these entries to be ± 1 in distinct rows, proving the claim. □

For $v = \operatorname{diag}(C)$, a common diagonal transport model is

$$\widehat{v}^{+} = (Q \odot Q)^{\top} v, \qquad v^{+} - \widehat{v}^{+} = \operatorname{diag}\Big(Q^{\top}[C - \operatorname{diag}(v)]Q\Big), \tag{59}$$

where $\odot$ denotes entrywise multiplication and $v^{+} = \operatorname{diag}(Q^{\top}CQ)$. The rule is exact if the old full moment is diagonal, but diagonal storage alone cannot verify that condition. Preserving the historical operator requires accounting for the omitted term or storing additional information.

Sign and permutation alignment can prevent state-reordering errors. It cannot recover discarded cross-coordinate moments under a genuine rotation, including a rotation within a repeated-eigenvalue subspace. This is the information limit behind basis transport; the index-retained recurrence in section 6.4 admits a convergence proof without reconstructing the historical operator.

7.5 From numerical error to optimization progress

The relevant numerical tolerance depends on the descent produced by the outer method. For a designated update D and realization $\widehat{D} = D + E$, Frobenius smoothness gives

$$\begin{aligned} F(W - \eta\widehat{D}) \le F(W) - \eta\,\langle \nabla F(W), D\rangle + \eta\,\|\nabla F(W)\|_F\,\|E\|_F \\ + \frac{L_F\eta^2}{2}\big(\|D\|_F + \|E\|_F\big)^2. \end{aligned} \tag{60}$$

Error can reduce alignment and increase update energy. If $\langle \nabla F, D\rangle \ge a\,\|\nabla F\|_F$ and $\|E\|_F \le e < a$, the linear descent coefficient remains at least $a - e$. An error bound is therefore most useful when measured against the available descent margin.

To combine different sources of error, one can telescope the outputs of a population target, structured fit, online estimate, stale representation, finite map, and distributed realization. The triangle inequality bounds their sum without an independence assumption, although the individual analyses may use different metrics. When exact polar is not a useful small-error reference, direct alignment analysis is preferable to (60). For the unregularized objective, decay belongs in the designated update as $D + \lambda W$.

Low precision adds another layer: roundoff can break symmetry or positive definiteness, while symmetrization, clipping, and extra damping alter the realized operator. Numerical analyses such as Polar Express and solver-focused systems work [29, 35] complement outer-loop theorems by controlling these effects. Their tolerances should reflect the update's descent margin and energy, not matrix-function accuracy alone.

8 Distributed realizations and structural fidelity

Distributed Shampoo, scalable Muon, NorMuon, and Dion support matrix-structured optimization at large scale [8, 26, 41, 45]. Distribution can preserve a logical update or change its statistical input, operator, or state. We use structural fidelity to distinguish these cases: after sharding and communication are specified, the update and the information it uses must still be identifiable.

8.1 Aggregation order changes the statistic

The first distinction arises before any matrix function is computed. Averaging local second moments is not the same as forming a second moment from the averaged gradient.

Proposition 8.1 (Local second moments differ from the global-mean second moment)**.** *For matrices G_j and nonnegative weights w_j summing to one, set $\overline{G} = \sum_j w_j G_j$. Then*

$$\sum_j w_j G_j G_j^\top - \overline{G}\,\overline{G}^\top = \sum_j w_j (G_j - \overline{G})(G_j - \overline{G})^\top \succeq 0. \tag{61}$$

Proof. Expand the right side and cancel the two cross terms using $\sum_j w_j G_j = \overline{G}$. □

The difference is a positive-semidefinite between-worker variation term, and the same identity holds for the right mode. On heterogeneous data this term can be large. Choosing either statistic is possible, but switching between them changes what the optimizer estimates.

A similar issue occurs after applying a nonlinear update map. For $G_1 = \mathrm{diag}(2, 1)$ and $G_2 = \mathrm{diag}(-1, 1)$,

$$\mathrm{polar}\left(\frac{G_1 + G_2}{2}\right) = I, \qquad \frac{\mathrm{polar}(G_1) + \mathrm{polar}(G_2)}{2} = \mathrm{diag}(0, 1). \tag{62}$$

Thus averaging locally orthogonalized momentum does not reproduce the update obtained by orthogonalizing global momentum.

8.2 An exact row-sharded polynomial construction

Nonlinearity does not mean that the entire matrix must be gathered onto one device. Some global maps can be evaluated through distributed sufficient products.

Proposition 8.2 (Exact row-sharded polynomial iteration)**.** *Partition X by rows as $X = (X_1^\top, \ldots, X_p^\top)^\top$. Define $H = X^\top X = \sum_j X_j^\top X_j$. For any fixed polynomial π, the block updates $X_j^+ = X_j \pi(H)$ jointly equal $X^+ = X\pi(X^\top X)$. Repeating this construction, with a newly aggregated Gram matrix at every inner step and identical scalar coefficients, implements the corresponding global polynomial iteration exactly in real arithmetic.*

Proof. The identity for H follows by block multiplication. Stacking the local outputs gives $X\pi(H)$. Applying the same argument inductively proves equality at each subsequent inner step. A global normalization is also exact if its norm is computed from the sum of local squared Frobenius norms. □

The quintic in (23) can be written as $X[aI + b(X^\top X) + c(X^\top X)^2]$, so this construction applies directly. Whether it is preferable to an all-gather depends on the matrix shape, shard orientation, and communication cost.

For an $m \times n$ row-sharded matrix on p workers, the construction replicates an $n \times n$ Gram matrix and needs $O(mn/p + n^2)$ update workspace per worker. Each inner step uses $O(mn^2/p)$ dense arithmetic for local Grams and right products; $X_j H^2$ can be evaluated as $(X_j H)H$ without forming

H^2. There is one Gram all-reduce per inner step and an initial scalar reduction for normalization. Under the usual uncompressed ring schedule, each worker moves approximately $2(p-1)n^2/p$ words per Gram. With per-message latency α and per-word cost β_{net}, the cost model is

$$2(p-1)\alpha + \frac{2(p-1)}{p} n^2 \beta_{\text{net}} \tag{63}$$

per Gram collective, ignoring overlap and topology effects. This is an execution model, not a communication lower bound or hardware benchmark. All-gather communicates on the scale of the matrix itself, so neither scheme dominates across all shapes and parallel layouts.

8.3 Storage and geometry partitioning

Sharding the storage of one logical matrix can preserve its geometry. Independent transforms on separate blocks generally do not. Replacing a full left factor by its block diagonal, for instance, removes cross-block correlations; a one-sided rectangular transform, blocked two-sided Shampoo, and a full transform with sharded state are distinct constructions.

Historical Shampoo factors make the difference concrete. The right Gram of a row-sharded gradient can be obtained by summation, whereas the full left Gram contains off-diagonal products $G_i G_j^\top$. Local $G_i G_i^\top$ alone cannot represent it. Assigning complete layer factors to different workers, by contrast, distributes the state without changing each layer's operator. A distributed implementation needs to specify which partition it uses [26].

Low-rank updates add a temporal aspect. Methods such as Dion can restrict the transformed subspace and carry residual state between steps [45, 46]. An error bound for the accumulated parameter update must track both omitted components and subspace evolution, not just the communicated rank. The same issue arises for subspace basis updates in Shampoo and SOAP [28].

8.4 Practical layouts and communication-reducing algorithms

The available information depends on the parallel layout. Data parallelism (DP) aggregates replicated gradient contributions; tensor parallelism (TP) splits a logical matrix along model dimensions; ZeRO-3 partitions parameters, gradients, and optimizer state, potentially across flattened buffers. A storage shard may therefore lack the matrix required by the optimizer. Reconstruction, communication of sufficient products, and shard-local transforms solve different problems.

Canzona decouples optimizer ownership from parameter sharding and pipelines TP reconstruction with load-balanced scheduling [56]. DMuon combines owner-based execution, shape-aware kernels, cost-based scheduling, and asynchronous parameter publication [57]. Both seek to change execution while preserving the update. Owners must hold complete matrices, state, and transform workspace, and unequal matrix shapes create unequal costs. Balancing parameter counts need not balance peak memory or optimizer work. Asynchronous execution can overlap computation and communication while respecting dependencies, without introducing stale gradients, momentum, or parameters.

MatrixFSDP makes whole-matrix ownership a ZeRO-3 placement, so backward reduction delivers each complete gradient to its owner [58]. Its communication-free claim covers optimizer-step

matrix collectives; forward/backward materialization and communication remain. The owner planner described in the cited version excludes TP fragments and therefore does not cover arbitrary combined TP/ZeRO-3 layouts. HyperParallel-FSDP instead uses layout-derived groups, deduplicated orthogonalization, and shape-fused Newton–Schulz in a topology-aware runtime [59]. Its results concern the studied hardware and layouts. In both cases, placement and scheduling must account for fragmented buffers, owner memory, overlap, and ownership-aware checkpointing.

Other methods reduce communication by changing the recurrence. MuonBP alternates shard-local and full-matrix orthogonalization with distinct step-size prescriptions [60]. EF21-Muon adds bidirectional compression and error-feedback state to a non-Euclidean linear-minimization-oracle method [61]. These estimators and updates need their own analysis. The placement of sign compression before or after the spectral oracle can also change whether descent is preserved, and output-side error feedback does not always repair it [62]. Results for an exact spectral oracle must further be distinguished from finite-polynomial implementations.

8.5 Assumptions for distributed guarantees

A distributed theorem first needs to identify the logical parameter blocks and aliases, aggregation order, and ownership of all historical state, including compression residuals. Collective products, synchronization and publication dependencies, parameter and gradient versions, and arithmetic precision then determine whether execution introduces rounding differences or changes the recurrence itself.

For an outer-loop proof with exact global matrix operations, proposition 8.2 supplies an algebraic implementation argument. Blocking, compression, delayed communication, or local normalization that changes the operator requires an additional approximation or descent bound. Time-to-target comparisons can then separate faster execution of the same update from a better cost–progress trade-off obtained by changing it.

9 Published empirical evidence and its interpretation

The six observations below test different parts of the preceding analysis: the timing of basis and scale updates, block scaling, polar accuracy, and execution cost. They come from separate studies rather than a common benchmark. Table 10 in appendix D records the source versions, experimental settings, and precise locations.

9.1 Separate timescales for bases and scales

SOAP's C4 experiment with a 660M decoder-only model provides evidence for separating basis refresh from scale adaptation. SOAP and Shampoo are close when the refresh interval is one. Both degrade as it increases to 100, but SOAP degrades more slowly [5, v2, Fig. 1 and Sections 5–6]. The recipe uses learning-rate and paired hyperparameter sweeps, mixed-bf16 training, fp32 factor updates, and QR-based basis refreshes after initialization. Coordinatewise scales continue to adapt between those refreshes.

Purifying Shampoo tests a related separation on Imagewoof with SimpleViT and ConvNeXt V2 trained for 100 epochs. EShampoo matches or improves on Adam-grafted Shampoo in final training cross-entropy while refreshing roots or bases every 100 steps and correcting eigenvalues every step [21, v1, Fig. 2 and Appendix E.1]. Ungrafted Shampoo matches the ViT result but not the ConvNeXt V2 result. These observations support separating basis refresh from scale correction, while showing that removing scale control remains architecture-dependent.

The experiments motivate distinct bounds for eigenbasis error, scale-estimation error, and state handling at a basis change. They do not isolate every aspect of historical-state transport: proposition 7.4 identifies an information limit, and section 6.4 establishes convergence for a specified recurrence.

9.2 Update scaling

Scalable Muon shows why the block scale in (19) must remain part of the comparison. For an approximately 800M dense model evaluated at 4B tokens of a 20B-token schedule, validation loss is 2.812 with baseline scaling, 2.789 with RMS normalization, and 2.789 with a shape-adjusted learning rate [8, v1, Table 1 and Section 3.1]. The recipe uses five Newton–Schulz steps and momentum 0.95. Panel-specific tuning, data, and precision fields remain unverified in the survey's record (table 10). Within this regime, the polar direction alone does not explain the result; the learning rate and update magnitude must be interpreted together.

9.3 Polar accuracy and training loss

Polar Express asks whether a more accurate polar calculation improves training. Its GPT-2-Small experiment on 1B FineWeb tokens varies inner depth from 2 to 30 and includes exact SVD [35, v5, Fig. 5 and Sections 4.2–4.3]. Five or six polynomial steps improve final validation loss over two or three. Further accuracy, including SVD, gives no additional loss improvement, while SVD roughly doubles training-step runtime, including forward and backward computation.

The ablation fixes learning rate 0.05 and weight decay 0.1, using four H100s, batch 32, mixed-bf16 training, and bf16 polynomial matrix-sign computation. It therefore identifies a particular setting in which more accurate matrix computation costs more without improving the reported loss. This is consistent with analyzing finite-map alignment and energy (lemma 6.2 and proposition 7.2), but does not prescribe a universal inner-iteration budget.

9.4 Iteration and wall-clock efficiency

The KL-Shampoo comparison makes the choice of efficiency measure decisive. With 450M Llama3, KL-Shampoo improves on Muon per iteration but not per wall-clock time; both improve on SOAP [10, v10, Fig. 9]. The panel uses 120 random-search runs per method and fp32 QR every ten iterations, without tuning refresh frequency for runtime. It should not be conflated with the paper's smaller-model, bf16 Figure 1 experiment using 150 runs per method.

DASH tests an execution improvement at matched reported quality. In its Llama-953M/C4 comparison, DIST and DASH both reach mean validation perplexity 11.87, while optimizer-step latency falls from 675 to 221 ms [29, v1, Table 1 and Section 5]. The matched row uses block

size 2,048, fp32 coupled-Newton inverse roots, refresh every step, and Frobenius normalization; the study uses three seeds and Adam grafting with a transferred learning rate. Unlike the Polar Express timing, this latency excludes forward and backward passes.

KL-Shampoo thus illustrates an iteration–execution trade-off, whereas DASH reduces optimizer time at the same reported quality. The numerical timings cannot be pooled across these studies: their training settings and timing denominators differ.

10 Open problems and predictive theory

The preceding comparisons explain how matrix-aware updates are constructed and which properties have been proved. A predictive theory must go further: it should identify when the retained information and chosen geometry improve quality enough to pay for their computation. Several concrete questions emerge from the analysis.

10.1 Statistical information and finite-time progress

In the frozen model, a preconditioner changes both initial-state contraction and orientation-weighted noise accumulation in (35). Minimizing the condition number alone does not optimize both terms. Minimizing only the stationary noise floor can instead favor an arbitrarily small effective step, at the expense of finite-time progress.

A natural target is to minimize a finite-time loss bound over a class such as $P = B \otimes A$, subject to fixed normalization and a state or communication budget. Normalization removes the redundancy between the scalar scale of P and the learning rate. Such a result would identify conditions on curvature, noise orientation, and initialization under which separable or one-sided transformations improve the bound. Extending it to historical estimates also requires controlling their feedback with the parameter trajectory.

10.2 Basis memory and scale memory

Diagonal storage poses a precise information limit: signed permutations are the only orthogonal changes that admit exact transport for every SPD moment (corollary 7.5). What can be recovered for restricted moment families, or approximated when rotations are controlled? The omitted term in (59) suggests studying diagonal-plus-low-rank states as a memory–error trade-off.

There is a separate optimization question. Generalized SOAP converges under a specified recurrence and schedule even with moving, current-sample-dependent bases [13], but its identity-basis specialization satisfies the same order of bound. The benefit of a learned basis and a particular state policy must therefore appear in constants or transient progress. An analysis should distinguish this benefit from preservation of the historical operator and handle repeated eigenspaces through projectors or full operators where appropriate. Policies tied to the base update, such as ARO, extend the question beyond eigenbasis tracking [12].

10.3 Inner accuracy and error allocation

When should an optimizer spend another matrix multiplication on a more accurate transform, and when should it refresh a stale estimate instead? Fixed-depth polar approximation is nonuniform near vanishing singular values, while damped inverse powers satisfy a damping-dependent operator Lipschitz bound (propositions 7.2 and 7.3). The appropriate accuracy criterion must therefore reflect the outer update's alignment and energy. Even an explicit dimension-free bound can be too conservative to select a useful iteration budget, as (55) illustrates.

A stopping principle could compare numerical, estimation, and temporal errors in compatible update norms, then allocate matrix products and synchronizations to the dominant contribution. Its guarantee would need to relate progress to gradient evaluations, matrix products, and communication. Low-precision positivity repairs and damping belong in this account because they change the realized operator.

10.4 Guarantees for practical schedules

Existing results address different parts of practical training. Finite-NS analyses retain empirical maps, generalized SOAP permits broad rotation choices, and AdamW-style Shampoo version 4 allows fixed second-moment EMA coefficients in a no-decay general-exponent regime [54]. Other quantities remain coupled. Fixed second-moment memory, fixed first-moment momentum, positive decay, and a long constant-learning-rate phase are not one interchangeable assumption.

A phase-wise treatment of warmup, sustained learning, and cooldown could clarify whether each phase approaches a noise-dependent neighborhood or yields a vanishing stationarity measure. For stale or grafted Shampoo, predictable-state estimates can be combined with positive metric variation in (69), without assuming cumulative monotonicity. Moving-basis results must identify the state recurrence they control. Last-iterate and parameter convergence then require arguments beyond average stationarity.

10.5 Distributed geometry and communication cost

The exact construction in proposition 8.2 shows that aggregated Gram products can implement a global polynomial. Its value depends on shape, shard orientation, collective latency, and replicated Gram size. Choosing the operator family together with its communication pattern raises a more selective question: which cross-block interactions deliver enough progress to justify retaining them?

Whole-matrix ownership and topology-aware layouts reduce reconstruction and synchronization costs while preserving a designated update [58, 59]. Choosing ownership, shard orientation, and overlap under peak-memory and dependency constraints remains open, as does predicting time to target. Optimizer-step communication savings have to be evaluated alongside communication in the rest of training.

For compressed updates, one could instead bound the accumulated effect of omitted directions or preserve a specified descent margin. This would provide a basis for comparing a cheaper modified transform with an exact, more expensive one. The state and arithmetic models in table 7 and the ring schedule in section 8 make the resource assumptions explicit.

10.6 Conditions for predictive comparisons

Table 5 collects conditions under which the preceding analysis already gives a useful conclusion, together with the quantities still needed for a performance comparison. A predictive account

Table 5: Conditional conclusions for structured optimization.

Condition or diagnostic	Conclusion available	Quantity still needed
Same instantaneous input, no damping or history	Two-sided fourth roots equal canonical polar	Historical state and momentum for trajectory comparison
One mode admits favorable curvature/noise traces and bounded radii	ASGO's bound isolates that mode's structural complexity	Actual radii, traces, and arithmetic cost
Fixed finite spectral map with positive scalar alignment bounds	Descent can be analyzed without uniform polar approximation	Tracking error and usable scalar constants
Slowly changing basis with rapidly changing scales	Separate refresh and scaling policies are meaningful design choices	State policy and measured or bounded update error
Row-sharded input with affordable replicated Gram	A global polynomial can be computed exactly in real arithmetic	Collective cost and rounding error

would combine these tests with measured spectra, noise orientation, matrix shape, and resource budgets. Generalization requires further assumptions: population risk also depends on the data and parameterization. Connecting matrix geometry to implicit bias therefore goes beyond the training-loss and stationarity results considered here.

11 Conclusion

Shampoo, SOAP, and Muon make different uses of matrix structure. Shampoo builds a separable preconditioner from historical statistics; SOAP separates a factorized basis from adaptive scales that need not be separable; Muon transforms the singular values of its update input. Their algebraic connections are informative only when the inputs, history, damping, and numerical maps are matched. Instantaneous preconditioning can equal polar even though historical variants follow different trajectories, and a finite spectral map can support descent without uniformly approximating polar.

These distinctions help explain why a more accurate statistic or matrix computation need not produce a better training outcome. In the frozen quadratic model, preconditioning changes both contraction and orientation-dependent noise accumulation. With moving bases, momentum, and estimated factors, the remaining task is to understand how those choices interact with the parameter trajectory and the cost of each update. Comparisons with AdamW should therefore report time to a specified quality together with memory use, tuning budget, precision, and the scope of the timing. That is the level at which the mathematical benefits of matrix structure and its practical costs can be judged together.

References

[1] Diederik P. Kingma and Jimmy Ba. Adam: A Method for Stochastic Optimization. In *International Conference on Learning Representations*, 2015. URL https://arxiv.org/abs/1412.6980.

[2] Ilya Loshchilov and Frank Hutter. Decoupled Weight Decay Regularization. In *International Conference on Learning Representations*, 2019. URL https://arxiv.org/abs/1711.05101.

[3] Vineet Gupta, Tomer Koren, and Yoram Singer. Shampoo: Preconditioned Stochastic Tensor Optimization. In *Proceedings of the 35th International Conference on Machine Learning*, volume 80 of *Proceedings of Machine Learning Research*, pages 1842–1850. PMLR, 2018. URL https://proceedings.mlr.press/v80/gupta18a.html.

[4] Rohan Anil, Vineet Gupta, Tomer Koren, Kevin Regan, and Yoram Singer. Scalable Second Order Optimization for Deep Learning. arXiv:2002.09018, 2020. URL https://arxiv.org/abs/2002.09018. Revised 2021.

[5] Nikhil Vyas, Depen Morwani, Rosie Zhao, Mujin Kwun, Itai Shapira, David Brandfonbrener, Lucas Janson, and Sham Kakade. SOAP: Improving and Stabilizing Shampoo using Adam. arXiv:2409.11321, 2024. URL https://arxiv.org/abs/2409.11321v2. Version 2, January 31, 2025; first preprint September 2024.

[6] Keller Jordan. Muon: An Optimizer for Hidden Layers in Neural Networks. Author blog, 2024. URL https://kellerjordan.github.io/posts/muon/. Published December 8, 2024; accessed October 7, 2026.

[7] Jeremy Bernstein and Laker Newhouse. Old Optimizer, New Norm: An Anthology. arXiv:2409.20325, 2024. URL https://arxiv.org/abs/2409.20325v1. Version 1.

[8] Jingyuan Liu, Jianlin Su, Xingcheng Yao, Zhejun Jiang, Guokun Lai, Yulun Du, Yidao Qin, Weixin Xu, Enzhe Lu, Junjie Yan, Yanru Chen, Huabin Zheng, Yibo Liu, Shaowei Liu, Bohong Yin, Weiran He, Han Zhu, Yuzhi Wang, Jianzhou Wang, Mengnan Dong, Zheng Zhang, Yongsheng Kang, Hao Zhang, Xinran Xu, Yutao Zhang, Yuxin Wu, Xinyu Zhou, and Zhilin Yang. Muon Is Scalable for LLM Training. arXiv:2502.16982, 2025. URL https://arxiv.org/abs/2502.16982v1. Version 1, February 24, 2025.

[9] Runa Eschenhagen, Anna Cai, Tsung-Hsien Lee, and Hao-Jun Michael Shi. Clarifying Shampoo: Adapting Spectral Descent to Stochasticity and the Parameter Trajectory. arXiv:2602.09314, 2026. URL https://arxiv.org/abs/2602.09314v1. Version 1, February 10, 2026.

[10] Wu Lin, Scott C. Lowe, Felix Dangel, Runa Eschenhagen, Zikun Xu, and Roger B. Grosse. Understanding and Improving Shampoo and SOAP via Kullback–Leibler Minimization. arXiv:2509.03378, 2026. URL https://arxiv.org/abs/2509.03378v10. Version 10, June 22, 2026; first preprint September 2025; ICLR 2026.

[11] Shuo Xie, Tianhao Wang, Beining Wu, and Zhiyuan Li. A Tale of Two Geometries: Adaptive Optimizers and Non-Euclidean Descent. arXiv:2511.20584, 2025. URL https://arxiv.org/abs/2511.20584v1. Version 1, November 25, 2025.

[12] Wenbo Gong, Javier Zazo, Qijun Luo, Puqian Wang, James Hensman, and Chao Ma. ARO: A New Lens On Matrix Optimization For Large Models. arXiv:2602.09006, 2026. URL https://arxiv.org/abs/2602.09006v1. Version 1, February 9, 2026.

[13] Yiwen Sun, Huan Li, and Zhouchen Lin. Convergence of Rotation-based Matrix Optimizers: A Unified Analysis of SOAP, Conda, and SPlus. arXiv:2609.24522, 2026. URL https://arxiv.org/abs/2609.24522v1. Version 1, September 21, 2026; subsumes arXiv:2604.21616.

[14] Ruoran Xu. Blog: Survey of Optimizers. arXiv:2608.28557v1, 2026. URL https://arxiv.org/abs/2608.28557v1. Preprint; accessed 7 October 2026.

[15] James Martens and Roger Grosse. Optimizing Neural Networks with Kronecker-factored Approximate Curvature. In *Proceedings of the 32nd International Conference on Machine Learning*, volume 37, pages 2408–2417. PMLR, 2015. URL https://proceedings.mlr.press/v37/martens15.html.

[16] Omead Pooladzandi and Xi-Lin Li. Curvature-Informed SGD via General Purpose Lie-Group Preconditioners. arXiv:2402.04553, 2024. URL https://arxiv.org/abs/2402.04553.

[17] John Duchi, Elad Hazan, and Yoram Singer. Adaptive Subgradient Methods for Online Learning and Stochastic Optimization. *Journal of Machine Learning Research*, 12(61):2121–2159, 2011. URL https://jmlr.org/papers/v12/duchi11a.html.

[18] Kang An, Yuxing Liu, Rui Pan, Yi Ren, Shiqian Ma, Donald Goldfarb, and Tong Zhang. ASGO: Adaptive Structured Gradient Optimization. arXiv:2503.20762, 2025. URL https://arxiv.org/abs/2503.20762v3. Version 3, October 29, 2025.

[19] Shuo Xie, Tianhao Wang, Sashank Reddi, Sanjiv Kumar, and Zhiyuan Li. Structured Preconditioners in Adaptive Optimization: A Unified Analysis. arXiv:2503.10537, 2025. URL https://arxiv.org/abs/2503.10537v2. Version 2, July 15, 2025.

[20] Depen Morwani, Itai Shapira, Nikhil Vyas, Eran Malach, Sham Kakade, and Lucas Janson. A New Perspective on Shampoo's Preconditioner. arXiv:2406.17748, 2024. URL https://arxiv.org/abs/2406.17748.

[21] Runa Eschenhagen, Aaron Defazio, Tsung-Hsien Lee, Richard E. Turner, and Hao-Jun Michael Shi. Purifying Shampoo: Investigating Shampoo's Heuristics by Decomposing its Preconditioner. arXiv:2506.03595, 2025. URL https://arxiv.org/abs/2506.03595v1. Version 1, June 4, 2025.

[22] Bing Liu, Wenjie Zhou, and Chengcheng Zhao. Rethinking Bregman Divergences in Kronecker-Factored Optimizers. arXiv:2606.00542, 2026. URL https://arxiv.org/abs/2606.00542.

[23] Vladimir Feinberg, Xinyi Chen, Y. Jennifer Sun, Rohan Anil, and Elad Hazan. Sketchy: Memory-efficient Adaptive Regularization with Frequent Directions. arXiv:2302.03764, 2023. URL https://arxiv.org/abs/2302.03764v2. Version 2.

[24] Wu Lin, Felix Dangel, Runa Eschenhagen, Juhan Bae, Richard E. Turner, and Alireza Makhzani. Can We Remove the Square-Root in Adaptive Gradient Methods? A Second-Order Perspective. arXiv:2402.03496, 2024. URL https://arxiv.org/abs/2402.03496v10. Version 10.

[25] Kevin Frans, Sergey Levine, and Pieter Abbeel. A Stable Whitening Optimizer for Efficient Neural Network Training. arXiv:2506.07254, 2025. URL https://arxiv.org/abs/2506.07254v3. Version 3, October 23, 2025.

[26] Hao-Jun Michael Shi, Tsung-Hsien Lee, Shintaro Iwasaki, Jose Gallego-Posada, Zhijing Li, Kaushik Rangadurai, Dheevatsa Mudigere, and Michael Rabbat. A Distributed Data-Parallel PyTorch Implementation of the Distributed Shampoo Optimizer for Training Neural Networks At-Scale. arXiv:2309.06497, 2023. URL https://arxiv.org/abs/2309.06497.

[27] Kyunghun Nam and Sumyeong Ahn. FOAM: Frequency and Operator Error-Based Adaptive Damping Method for Reducing Staleness-Oriented Error for Shampoo. arXiv:2606.02365, 2026. URL https://arxiv.org/abs/2606.02365.

[28] Alan Milligan, Zikun Xu, Simon Lacoste-Julien, Felix Dangel, and Wu Lin. Reparametrizing Shampoo and SOAP for Subspace Basis Updates and BFloat16 Storage. arXiv:2605.26327, 2026. URL https://arxiv.org/abs/2605.26327.

[29] Ionut-Vlad Modoranu, Philip Zmushko, Erik Schultheis, Mher Safaryan, and Dan Alistarh. DASH: Faster Shampoo via Batched Block Preconditioning and Efficient Inverse-Root Solvers. arXiv:2602.02016, 2026. URL https://arxiv.org/abs/2602.02016v1. Version 1, February 2026.

[30] Sike Wang, Pan Zhou, Jia Li, and Hua Huang. 4-bit Shampoo for Memory-Efficient Network Training. arXiv:2405.18144, 2024. URL https://arxiv.org/abs/2405.18144v3. Version 3.

[31] Haonan Wang, Yu Wu, Minghui Liwang, Xinlei Yi, and Yiguang Hong. Convergence of Practical Muon. arXiv:2609.33152, 2026. URL https://arxiv.org/abs/2609.33152v1. Version 1, September 27, 2026.

[32] Gyu Yeol Kim and Min-hwan Oh. Convergence of Muon with Newton–Schulz. arXiv:2601.19156, 2026. URL https://arxiv.org/abs/2601.19156v1. Version 1, January 27, 2026; accepted at ICLR 2026.

[33] Hanyang Peng, Hui Wang, and Yue Yu. Convergence of Practical Muon with Finite Newton–Schulz Iterations and Nesterov Momentum. arXiv:2609.39595, 2026. URL https://arxiv.org/abs/2609.39595v1. Version 1, September 30, 2026.

[34] Da Chang, Yongxiang Liu, Huihui Zhou, and Ganzhao Yuan. On the Convergence of Muon and Beyond. arXiv:2509.15816, 2026. URL https://arxiv.org/abs/2509.15816v6. Version 6, September 30, 2026; first preprint September 19, 2025.

[35] Noah Amsel, David Persson, Christopher Musco, and Robert M. Gower. The Polar Express: Optimal Matrix Sign Methods and Their Application to the Muon Algorithm. arXiv:2505.16932, 2026. URL https://arxiv.org/abs/2505.16932v5. Version 5, May 4, 2026; first preprint May 22, 2025; ICLR 2026.

[36] Minxin Zhang, Yuxuan Liu, and Hayden Schaeffer. AdaGrad Meets Muon: Adaptive Stepsizes for Orthogonal Updates. arXiv:2509.02981, 2025. URL https://arxiv.org/abs/2509.02981v2. Version 2.

[37] Jingru Li, Yibo Fan, and Huan Li. Variance-Adaptive Muon: Pre-Orthogonalization Variance Modulation for Efficient Language Model Pretraining. arXiv:2601.14603, 2026. URL https://arxiv.org/abs/2601.14603v2. Version 2.

[38] Ziyue Liu, Ruijie Zhang, Zhengyang Wang, Yequan Zhao, Yupeng Su, Zi Yang, and Zheng Zhang. Muon^2: Boosting Muon via Adaptive Second-Moment Preconditioning. arXiv:2604.09967, 2026. URL https://arxiv.org/abs/2604.09967v2. Version 2.

[39] Xianbiao Qi, Marco Chen, Jiaquan Ye, Yelin He, and Rong Xiao. Delving into Muon and Beyond: Deep Analysis and Extensions. arXiv:2602.04669, 2026. URL https://arxiv.org/abs/2602.04669v1. Version 1.

[40] Chongjie Si, Debing Zhang, and Wei Shen. AdaMuon: Adaptive Muon Optimizer. arXiv:2507.11005, 2025. URL https://arxiv.org/abs/2507.11005v3. Version 3, December 24, 2025.

[41] Zichong Li, Liming Liu, Chen Liang, Weizhu Chen, and Tuo Zhao. NorMuon: Making Muon More Efficient and Scalable. arXiv:2510.05491, 2025. URL https://arxiv.org/abs/2510.05491v1. Version 1, October 7, 2025.

[42] Yechen Zhang, Shuhao Xing, Junhao Huang, Kai Lv, Yunhua Zhou, Xipeng Qiu, Qipeng Guo, and Kai Chen. Mousse: Rectifying the Geometry of Muon with Curvature-Aware Preconditioning. arXiv:2603.09697, 2026. URL https://arxiv.org/abs/2603.09697v2. Version 2, April 1, 2026.

[43] Kimi Team. Kimi K2: Open Agentic Intelligence. arXiv:2507.20534, 2026. URL https://arxiv.org/abs/2507.20534v2. Version 2, February 3, 2026; first preprint July 28, 2025; group authorship as listed by the report.

[44] Arash Lagzian, Paniz Halvachi, Junming Zhang, Zhouhan Lin, and Dianbo Liu. AF-Muon: An AdamW-Free Muon Optimizer for Tied-Embedding Models. arXiv:2610.01395, 2026. URL https://arxiv.org/abs/2610.01395v1. Version 1, October 1, 2026.

[45] Kwangjun Ahn, Byron Xu, Natalie Abreu, Ying Fan, Gagik Magakyan, Pratyusha Sharma, Zheng Zhan, and John Langford. Dion: Distributed Orthonormalized Updates. arXiv:2504.05295, 2025. URL https://arxiv.org/abs/2504.05295v3. Version 3, September 15, 2025; first preprint April 7, 2025.

[46] Noah Amsel, Jack Zhang, Kwangjun Ahn, Ali Naeimi, Austin Feng, Berlin Chen, Tri Dao, and John Langford. Dion3: Full-Stack Orthogonal Updates. arXiv:2608.11612, 2026. URL https://arxiv.org/abs/2608.11612v1. Version 1, August 12, 2026.

[47] Liming Liu, Zhenghao Xu, Zixuan Zhang, Hao Kang, Zichong Li, Chen Liang, Weizhu Chen, and Tuo Zhao. COSMOS: A Hybrid Adaptive Optimizer for Memory-Efficient Training of LLMs. arXiv:2502.17410, 2025. URL https://arxiv.org/abs/2502.17410v3. Version 3.

[48] Ruijie Zhang, Yequan Zhao, Ziyue Liu, Zhengyang Wang, Dongyang Li, Yupeng Su, Sijia Liu, and Zheng Zhang. TEON: Tensorized Orthonormalization Beyond Layer-Wise Muon for Large Language Model Pre-Training. arXiv:2601.23261, 2026. URL https://arxiv.org/abs/2601.23261v2. Version 2.

[49] Arman Bolatov, Artem Riabinin, Nikita Kornilov, Andrey Veprikov, Samuel Horváth, Martin Takáč, and Aleksandr Beznosikov. LionMuon: Alternating Spectral and Sign Descent for Efficient Training. arXiv:2605.19811, 2026. URL https://arxiv.org/abs/2605.19811v3. Version 3.

[50] Feihu Huang, Yuning Luo, and Songcan Chen. LiMuon: Light and Fast Muon Optimizer for Large Models. arXiv:2509.14562, 2025. URL https://arxiv.org/abs/2509.14562v5. Version 5.

[51] Jiaxiang Li and Mingyi Hong. A Note on the Convergence of Muon. arXiv:2502.02900, 2025. URL https://arxiv.org/abs/2502.02900v2. Version 2, June 1, 2025.

[52] Arthur C. B. de Oliveira, Dhruv D. Jatkar, Guilherme S. Vicinansa, and Eduardo D. Sontag. Convergence Guarantees for Muon: New Parameter Regimes and Generalizations. arXiv:2609.30546, 2026. URL https://arxiv.org/abs/2609.30546v1. Version 1, September 24, 2026.

[53] Thang Do, Steffen Dereich, and Arnulf Jentzen. On MUON Optimization: From Non-Convergence to an Error Analysis with Polar Express and the Newton–Schulz Polynomial from Implementations. arXiv:2608.04607, 2026. URL https://arxiv.org/abs/2608.04607v1. Version 1, August 5, 2026.

[54] Huan Li, Yiming Dong, and Zhouchen Lin. Convergence Rate Analysis of the AdamW-Style Shampoo: Unifying One-sided and Two-Sided Preconditioning. arXiv:2601.07326, 2026. URL https://arxiv.org/abs/2601.07326v4. Version 4, June 14, 2026; version 3 is the ICML 2026 camera-ready.

[55] Thibaut Boissin, Thomas Massena, Franck Mamalet, and Mathieu Serrurier. Turbo-Muon: Almost-Orthogonal Pre-Conditioning for Fast Muon Updates. arXiv:2512.04632, 2025. URL https://arxiv.org/abs/2512.04632v2. Version 2.

[56] Liangyu Wang, Siqi Zhang, Junjie Wang, Yiming Dong, Bo Zheng, Zihan Qiu, Shengkun Tang, Di Wang, Rui Men, and Dayiheng Liu. Canzona: A Unified, Asynchronous, and Load-Balanced Framework for Distributed Matrix-based Optimizers. arXiv:2602.06079, 2026. URL https://arxiv.org/abs/2602.06079v1. Version 1.

[57] Vincent Chen, Starrick Liu, Regis Cheng, Dance Yang, Shalfun Li, Ryan Yu, Lucy Liang, Hang Su, Roy Gan, Hao Wang, and Qian Wang. DMuon: Efficient Distributed Muon Training with Near-Adam Overhead. arXiv:2606.27153, 2026. URL https://arxiv.org/abs/2606.27153v1. Version 1.

[58] Ming Gao, Yanwu Xu, and Hao Zhang. MatrixFSDP: communication-free matrix optimizers under ZeRO-3 parameter sharding. arXiv:2607.05895, 2026. URL https://arxiv.org/abs/2607.05895v2. Version 2.

[59] Mo Sun, Yifan Yao, Yanwei Liu, Luobin Liu, Zhenzhang Yang, Kaisheng Wang, Xiangyu Meng, Chen Li, Xizheng Pang, Huilan Li, Xinglei Xu, Yushi Cui, Xinyao Lin, Kaiqi Chen, Jie Zhang, Zeke Wang, and Teng Su. HyperParallel-FSDP: Topology-Aware Fully Sharded Training with Layout-Driven Muon

on Ascend SuperPods. arXiv:2609.21594, 2026. URL https://arxiv.org/abs/2609.21594v1. Version 1.

[60] Ahmed Khaled, Kaan Ozkara, Tao Yu, Mingyi Hong, and Youngsuk Park. MuonBP: Faster Muon via Block-Periodic Orthogonalization. arXiv:2510.16981, 2025. URL https://arxiv.org/abs/2510.16981v1. Version 1.

[61] Kaja Gruntkowska, Alexander Gaponov, Zhirayr Tovmasyan, and Peter Richtárik. Error Feedback for Muon and Friends. arXiv:2510.00643, 2025. URL https://arxiv.org/abs/2510.00643v1. Version 1.

[62] Maria Smirnova and Alexey Kravatskiy. Sign compression for Muon: SignMuon, MuonSign, and the Limits of Error Feedback. arXiv:2607.29674, 2026. URL https://arxiv.org/abs/2607.29674v1. Version 1.

A Proof details and auxiliary identities

A.1 The cumulative Shampoo potential

We give the matrix case of Gupta et al. [3, Theorem 7] in the notation of theorem 6.1. All matrices are positive definite after initialization. Write $x_t = \mathrm{vec}(W_t)$, $u = \mathrm{vec}(W_*)$, and $g_t = \mathrm{vec}(G_t)$. The update $x_{t+1} = x_t - \eta H_t^{-1} g_t$, with $H_t = R_t^{1/4} \otimes L_t^{1/4}$, satisfies

$$\langle g_t, x_t - u\rangle \le \frac{\|x_t - u\|_{H_t}^2 - \|x_{t+1} - u\|_{H_t}^2}{2\eta} + \frac{\eta}{2} g_t^\top H_t^{-1} g_t. \tag{64}$$

The inequality also holds with H_t-metric projection onto a convex set containing u. Here $\|v\|_H^2 = v^\top H v$. Summing the numerator leaves initial and negative terminal terms, together with $\sum_{t=2}^T \|x_t - u\|_{H_t - H_{t-1}}^2$. Cumulative factors, their $1/4$ powers, and the Kronecker product increase in Loewner order, giving $H_t \succeq H_{t-1}$. If $\|x_t - u\|_2 \le D$, the displacement contribution is at most $D^2 \operatorname{tr}(H_T)/(2\eta)$.

The rank-sensitive inequality is

$$\frac{1}{r} g g^\top \preceq I_n \otimes G G^\top, \qquad \frac{1}{r} g g^\top \preceq G^\top G \otimes I_m, \quad r \ge \operatorname{rank}(G). \tag{65}$$

For the first inequality, g lies in the range of $B = I_n \otimes G G^\top$ and $g^\top B^\dagger g = \operatorname{tr}[G^\top (G G^\top)^\dagger G] = \operatorname{rank}(G)$. Cauchy–Schwarz applied to $\left\langle B^{1/2} v, B^{\dagger/2} g \right\rangle$ gives $\langle v, g\rangle^2 \le r v^\top B v$. Transposition proves the other inequality. Support pseudoinverses are used for this bound; the algorithmic preconditioners remain damped and invertible.

Let $S_t = \varepsilon I + (1/r)\sum_{s\le t} g_s g_s^\top$, $A_t = I_n \otimes L_t$, and $B_t = R_t \otimes I_m$. Equation (65) implies $S_t \preceq A_t, B_t$. For SPD matrices, the geometric mean is $A \# B = A^{1/2}(A^{-1/2} B A^{-1/2})^{1/2} A^{1/2}$. Its monotonicity gives $S_t = S_t \# S_t \preceq A_t \# B_t$. Since the Kronecker factors commute, $A_t \# B_t = R_t^{1/2} \otimes L_t^{1/2} = H_t^2$. Square-root monotonicity and inverse order then yield $S_t^{1/2} \preceq H_t$ and $H_t^{-1} \preceq S_t^{-1/2}$.

The scalar-valued function $X \mapsto \operatorname{tr}(X^{1/2})$ is concave on SPD matrices, with differential $\frac{1}{2}\operatorname{tr}(X^{-1/2}\, dX)$. Its supporting-hyperplane inequality at S_t is

$$\operatorname{tr}(S_t^{1/2}) - \operatorname{tr}(S_{t-1}^{1/2}) \ge \tfrac{1}{2}\operatorname{tr}[S_t^{-1/2}(S_t - S_{t-1})]. \tag{66}$$

Therefore,

$$\sum_{t=1}^T g_t^\top H_t^{-1} g_t \le r \sum_{t=1}^T \operatorname{tr}[S_t^{-1/2}(S_t - S_{t-1})] \tag{67}$$

$$\le 2r \operatorname{tr}(S_T^{1/2} - S_0^{1/2}) \le 2r \operatorname{tr}(H_T). \tag{68}$$

Convexity gives $f_t(W_t) - f_t(W_*) \le \langle G_t, W_t - W_*\rangle$. Combining this with (64) and $\operatorname{tr}(H_T) = \operatorname{tr}(R_T^{1/4}) \operatorname{tr}(L_T^{1/4})$ proves the regret bound and its optimized stepsize.

Nonmonotone metrics. For an arbitrary SPD sequence, the displacement contribution is bounded by

$$\frac{D^2}{2\eta}\left[\operatorname{tr}(H_1) + \sum_{t=2}^T \operatorname{tr}\big((H_t - H_{t-1})_+\big)\right], \tag{69}$$

where A_+ is the positive spectral part of a symmetric matrix. The bound follows from $v^\top A v \le D^2 \operatorname{tr}(A_+)$ for $\|v\|_2 \le D$. EMA, basis changes, and approximate factors therefore require control of positive metric variation, not only the endpoint metric.

A.2 Separable moments and the population KL objective

Let $C = \mathbb{E}[\operatorname{vec}(G)\operatorname{vec}(G)^\top] \succ 0$ be finite, as in (17). Interpreting this possibly uncentered moment as an auxiliary Gaussian covariance imposes no zero-mean assumption on G. For SPD $A \in \mathbb{R}^{m\times m}$ and $B \in \mathbb{R}^{n\times n}$, the Gaussian KL objective depends on $B \otimes A$ through, up to a factor $1/2$,

$$J(A,B) = n \log\det A + m \log\det B + \mathbb{E}\operatorname{tr}(A^{-1} G B^{-1} G^\top). \tag{70}$$

Column-major vectorization gives the trace identity. Setting the A differential to zero yields $nA^{-1} - A^{-1}\mathbb{E}[GB^{-1}G^\top]A^{-1} = 0$, and likewise for B. Any interior stationary point consequently satisfies

$$A = \frac{1}{n}\mathbb{E}[GB^{-1}G^\top], \qquad B = \frac{1}{m}\mathbb{E}[G^\top A^{-1} G]. \tag{71}$$

The transformation $(A,B) \mapsto (cA, B/c)$ preserves both covariance and objective; normalization fixes this gauge. These equations characterize population stationarity, not online convergence.

For singular C, the finite expression remains a Gaussian cross-entropy fitting objective up to constants and scale. The KL divergence is infinite, however, because the singular Gaussian occupies a proper subspace of zero probability under the SPD Gaussian. It cannot be recovered by discarding a finite constant. An interior minimizer may require regularization, and damping changes the objective or fixed-point map; see Lin et al. [10].

When $C = B \otimes A$ exactly, its (j,k) block is $B_{jk}A$. Summing diagonal blocks proves $\mathbb{E}[GG^\top] = \operatorname{tr}(B)A$; the transposed contraction gives $\mathbb{E}[G^\top G] = \operatorname{tr}(A)B$. This proves the covariance-scale correction in (14) under exact separability. Arbitrary mode-wise Gram matrices need not factorize the covariance.

A.3 Resolvent proof for damped inverse powers

To prove proposition 7.3, no commutativity between A and B is needed. For $0 < p < 1$, spectral calculus gives

$$(A+\delta I)^{-p} = \frac{\sin(\pi p)}{\pi}\int_0^\infty t^{-p}(A + (\delta+t)I)^{-1}\,dt. \tag{72}$$

For $s = \delta + t > 0$, the resolvent identity is

$$(A+sI)^{-1} - (B+sI)^{-1} = (A+sI)^{-1}(B-A)(B+sI)^{-1}. \tag{73}$$

Both inverse factors have operator norm at most s^{-1}. The integral converges in operator norm, and hence

$$\left\|(A+\delta I)^{-p} - (B+\delta I)^{-p}\right\|_{\mathrm{op}} \le \frac{\sin(\pi p)}{\pi}\|A-B\|_{\mathrm{op}}\int_0^\infty \frac{t^{-p}}{(\delta+t)^2}\,dt \tag{74}$$

$$= p\delta^{-p-1}\|A-B\|_{\mathrm{op}}. \tag{75}$$

The last integral is $\delta^{-p-1}\Gamma(1-p)\Gamma(1+p)$, and the reflection identity gives the constant. For $p = 1$, the resolvent identity with $s = \delta$ proves the result directly.

A.4 A factorized basis need not define a separable operator

Let $Q = Q_R \otimes Q_L$ and $P = Q\,\mathrm{diag}(\mathrm{vec}(S))Q^\top$, with every $S_{ij} > 0$. In this fixed basis, P is a single SPD Kronecker product if and only if $S = ab^\top$ for positive vectors a, b. Sufficiency follows from $\mathrm{diag}(\mathrm{vec}(ab^\top)) = \mathrm{diag}(b) \otimes \mathrm{diag}(a)$. For necessity, transform a putative SPD factorization into the Q basis. If $\widetilde{B} \otimes \widetilde{A}$ is diagonal and both factors are SPD, their positive diagonal entries force every off-diagonal entry of each factor to vanish. The remaining entries factor as $S_{ij} = \widetilde{A}_{ii}\widetilde{B}_{jj}$. General elementwise adaptive scales need not satisfy this rank-one condition, as in SOAP (section 2).

B Theorem statements and assumptions

The bounds below use the survey's column-major and norm conventions. Each entry records the source version, assumptions, and parameter regime needed to interpret the result.

B.1 Cumulative Shampoo: the original regret certificate

Source. Gupta et al. [3], ICML 2018, Theorem 7 and its proof, PDF pp. 4–5. **Algorithm and assumptions.** Cumulative two-sided Gram matrices, exact fourth roots, convex losses, a uniform rank bound r, and Frobenius distance at most D to the comparator. A constant stepsize $D/\sqrt{2r}$ gives (40). Bounded iterates must be assumed or enforced by the appropriate metric projection. **Mechanism and scope.** Rank domination, monotone metric potentials, and trace concavity are derived in appendix A.1. The result excludes EMA, momentum, arbitrary grafting, intermittent roots, and finite precision. The rank bound concerns each gradient, not a population covariance.

B.2 One-sided structure in the unified adaptive analysis

Source. Xie et al. [19], arXiv:2503.10537v2, Theorem 4.2 and Algorithm 4, PDF p. 11 for the theorem; Theorem 4.4 gives a related convex stochastic specialization. **Algorithm.** The normalized left statistic is $L_t = \varepsilon I + (1/n)\sum_{s\le t} G_s G_s^\top$, and the variable metric is $I_n \otimes L_t^{1/2}$ in this survey's vectorization. The update projects in that metric onto the specified matrix-norm domain. **Bound.** With comparator radius D_{op} and the source normalization,

$$\mathcal{R}_T \le \left(\frac{D_{\mathrm{op}}^2}{2n\eta} + \eta\right)\left[\sqrt{n}\ \mathrm{tr}\left(\left[\sum_{t=1}^{T} G_t G_t^\top\right]^{1/2}\right) + mn\sqrt{\varepsilon}\right]. \tag{76}$$

For $\|W\|_{\mathrm{op}} \le R$, taking $D_{\mathrm{op}} \le 2R$ and $\eta = \sqrt{2/n}R$ gives leading term $2\sqrt{2}R\,\mathrm{tr}[(\sum_t G_t G_t^\top)^{1/2}]$, plus initialization. Equation (42) retains this factor normalization, on which its constants depend. **Scope.** The framework uses structured matrix algebras. Two-sided Kronecker products do not form a linear matrix algebra, so a separable implementation needs a separate check of the hypotheses.

B.3 ASGO: curvature and noise in a selected mode

Source. An et al. [18], arXiv:2503.20762v3, Theorem 3, PDF p. 5. **Algorithm.** The one-sided accumulator starts at zero and sums $G_t G_t^\top$. Its metric adds εI after taking the square root, unlike

$(S_t + \varepsilon I)^{1/2}$. The stepsize is $\eta = D_{\mathrm{op}}$. **Assumptions and conclusion.** The objective is convex and smooth with respect to $\|X\|_A^2 = \mathrm{tr}(X^\top AX)$. Unbiased independent minibatch averaging reduces the single-sample matrix noise envelope by B. Finite spectral and Frobenius iterate radii give (43); dimension or architecture comparisons must retain those radii and $\mathrm{tr}(A), \mathrm{tr}(V)$. **Scope.** The guarantee is one-sided and does not cover two-sided Shampoo or polar updates. A horizon-independent radius for an unprojected trajectory requires further assumptions.

B.4 AdamW-style Shampoo with coupled parameters

Source. Li et al. [54], arXiv:2601.07326v4, Theorem 1 in Section 2 and Theorem 2 in Section 3. Theorem 1 retains the principal bound of v1's Theorem 2.1. **Algorithm.** Equation (44), with exact current factors, $p, q \in [1, \infty]$, $p^{-1} + q^{-1} = 1$, zero momentum/statistic initialization, and decoupled decay. **Noise and regularity.** The objective is lower bounded and L_F-smooth. Conditional noise $N_t = G_t - \nabla F(W_t)$ is unbiased and satisfies $\mathbb{E}[N_t N_t^\top \mid \mathcal{F}_{t-1}] \preceq \Sigma_L$ and $\mathbb{E}[N_t^\top N_t \mid \mathcal{F}_{t-1}] \preceq \Sigma_R$. Set $\sigma^2 = (\mathrm{tr}\,\Sigma_L + \mathrm{tr}\,\Sigma_R)/2$. The damped left and right statistics must be at least $\widehat{\varepsilon} I$ in Loewner order, with $\widehat{\varepsilon} \geq \varepsilon > 0$. **Schedule.** For $\gamma \in (0, 1)$, $\tau \in (0, 1]$, and $\widehat{\sigma}^2 = \max\{\sigma^2, L_F\Delta/(T\gamma^2)\}$, Theorem 1 prescribes

$$1 - \theta = \sqrt{\frac{L_F \Delta}{T\widehat{\sigma}^2}}, \qquad \theta \leq \beta \leq \sqrt{\theta}, \qquad \varepsilon = \frac{\tau \widehat{\sigma}^2}{m+n}, \tag{77}$$

$$\eta = \sqrt{\frac{\widehat{\varepsilon}\Delta}{4L_F T\widehat{\sigma}^2}}, \qquad 0 \leq \lambda \leq \frac{1}{\sqrt{1152\widehat{\varepsilon}}\, T^{3/4}} \left(\frac{L_F^3 \widehat{\sigma}^2}{\Delta}\right)^{1/4}. \tag{78}$$

It also requires $\|W_1\|_{\mathrm{op}} \leq \sqrt{\widehat{\varepsilon} T\Delta/(L_F\widehat{\sigma}^2)}$. **Conclusion and scope.** The average nuclear-gradient bound is (45). The universal lower bound $\widehat{\varepsilon} = \varepsilon$ gives a prefactor at most $127\sqrt{(m+n)/\tau}$. Any choice of $\widehat{\varepsilon}$ must lower-bound the actual statistics. Arbitrary fixed hyperparameters, cached inverse powers, finite precision, and compressed state are outside the result.

General-exponent extension. Theorem 2 allows $p, q \in (0, \infty]$, $p^{-1} + q^{-1} = 2 - \omega$ with $0 < \omega \leq 2$, but sets $\lambda = 0$. It retains the displayed θ, ε, and $\widehat{\sigma}$ choices, takes any fixed $\beta \in (0, 1)$ independent of T, and uses

$$\eta = \widehat{\varepsilon}^{1-\omega/2} \sqrt{\frac{\Delta}{4L_F T\widehat{\sigma}^2}}. \tag{79}$$

The power measure $\Psi_\omega(G) = \sum_i \sigma_i(G)^\omega$ is the source's $\||G|^\omega\|_*$, not an entrywise power. Its average bound is $\mathcal{O}(T^{-\omega/4} + T^{-1/2})$ in the positive-noise regime, with constants depending on dimension, damping, noise, ω, and fixed β. At $\omega = 2$, the update reduces to momentum SGD. The extension permits a fixed second-moment EMA coefficient but requires zero decay; first-moment momentum remains horizon-dependent.

B.5 Convergent Taylor-family NS for Muon

Source. Kim and Oh [32], arXiv:2601.19156v1, Definition 2 and Theorems 1–2, PDF pp. 5–6; empirical-polynomial limitation in Appendix H.4. **Algorithm and geometry.** The source uses

$M_t = \beta M_{t-1} + G_t$ and $X_{t,0} = M_t / \max\{1, \|M_t\|_F\}$ before a finite Taylor-family inverse-square-root iteration. For order κ,

$$p_\kappa(\lambda) = \sum_{j=0}^{\kappa} \frac{\binom{2j}{j}}{4^j}(1-\lambda)^j, \qquad X_{k+1} = p_\kappa(X_k X_k^\top) X_k. \tag{80}$$

The proof uses the theorem's normalization, support residual, and contraction of this NS family. The objective satisfies operator-to-nuclear smoothness; the oracle is unbiased with bounded variance and uses minibatch averaging. **Schedule and bound.** In the nondegenerate stochastic regime, $1-\beta = \min\{\sqrt{L_{\mathrm{op}}\Delta B}/(\sigma\sqrt{rT}), 1\}$ and $\eta = \sqrt{(1-\beta)\Delta/(TL_{\mathrm{op}})}$, with the source momentum convention. Equation (49) contains a finite-iteration factor depending on the initialized support residual. Making it uniform along the outer trajectory requires a residual uniformly below one. Exact polar removes this factor. **Scope.** The empirical quintic $(3.4445, -4.775, 2.0315)$ lies outside the Taylor family. Its finite-step map, weight decay, auxiliary optimizers, and finite precision need separate arguments.

B.6 Practical five-step Muon with decay

Source. Wang et al. [31], arXiv:2609.33152v1, Theorem 1, PDF p. 6; constants are defined in its equations (11) and (20). **Algorithm.** The mathematical update retains the empirical five-step quintic, Nesterov-style momentum input, normalization floor, and decoupled decay. Assumptions are Frobenius smoothness, unbiased bounded-variance minibatch gradients, and exact arithmetic. **Schedule.** With the source's positive constants c_1, c_2, c_3, C_2, C_3,

$$\eta = c_1 T^{-3/4}, \quad 1-\beta = c_2 T^{-1/2}, \quad 0 < \varepsilon \le c_3 T^{-1/4}, \quad \lambda = \frac{C_2}{2(\|W_0\|_F + C_3 c_1 T^{1/4})}. \tag{81}$$

Decay must satisfy the last relation. The stochastic bound is $\mathcal{O}((L_F \Delta\sigma^2/B)^{1/4} T^{-1/4})$ in the positive-noise regime; its constants do not cover degenerate zero-noise cases by substitution. **Scope.** The theorem analyzes the empirical polynomial under a coupled schedule in exact arithmetic. Matrix dimensions and coefficient estimates remain in the constants. Fixed schedules and low precision require further analysis.

B.7 Layer-wise finite NS and Nesterov tracking

Source. Peng et al. [33], arXiv:2609.39595v1, Theorem 3.4 and Corollary 3.5, PDF pp. 8–9. **Algorithm.** At the same full parameter state, each block uses its own Nesterov momentum input and fixed-depth empirical NS map, with Frobenius normalization and a separate zero-input case. The update is in exact arithmetic, without weight decay or auxiliary AdamW for other parameter classes. **Assumptions.** The objective is lower bounded, with blockwise generalized smoothness and a bounded-variance stochastic oracle. Blocks need not be independent, nor the objective separable. Frobenius smoothness is a specialization of the setting. **Schedule and conclusion.** Writing $\delta = 1-\beta_1$, the corollary takes $\delta = \delta_0 T^{-1/2}$, $\eta = \eta_0 T^{-3/4}$, and $\beta_2 = 1 - \delta^a$ for $a > 1/2$. For fixed block count, scalar-map, noise, and regularity constants, the average sum of blockwise Frobenius gradient norms is $\mathcal{O}(T^{-1/4})$. Matching the momentum coefficients corresponds to $a = 1$. **Mechanism and scope.** The finite map has positive alignment and bounded energy, with tracking

error controlled by the coupled schedules. No uniform exact-polar approximation is required. The estimate has no explicit dimension/rank factor, but a width-uniform claim needs control of the other constants. It supplies no memory-cost bound.

B.8 Variance-reduced Muon: a different oracle

Source. Chang et al. [34], arXiv:2509.15816v6, Theorems 6 and 8, PDF pp. 9–10; additional PL results impose further assumptions. **MVR1.** The schedule is $\eta_t = t^{-3/4}$ and $\beta_t = 1 - t^{-1/2}$, with the specified bounded decay. The expected average Frobenius-gradient rate is $\widetilde{\mathcal{O}}(T^{-1/4})$, where the tilde includes factors such as $\log T$. **MVR2.** The schedule is $\eta_t = t^{-2/3}$ and $\beta_t = 1 - t^{-2/3}$, with the corresponding decay restriction. The rate is $\widetilde{\mathcal{O}}(T^{-1/3})$ under stronger assumptions, including mean-square smoothness. Each correction evaluates the same fresh sample at the current and previous iterates, rather than reusing a stored noisy gradient. **Comparability.** Comparisons with single-evaluation Muon must count gradient evaluations and retain the sample-reuse and oracle-smoothness requirements. Faster PL conclusions additionally require gradient bounds and parameter restrictions beyond nonconvex stationarity.

B.9 Soft polar and asymptotic guarantees

Source. de Oliveira et al. [52], arXiv:2609.30546v1, Theorem 1 and Corollary 1, followed by the soft-Muon specialization and boundedness analysis. **Object and assumptions.** The deterministic heavy-ball-type recursion uses uniformly bounded SPD preconditioners and restricted step/momentum parameters. The objective satisfies smoothness and properness/bounded-sublevel conditions. For soft polar, the left matrix is $(MM^\top + \varepsilon^2 I)^{1/2}$. The positive floor gives a lower bound, while an upper bound needs momentum control. **Conclusion and scope.** The Lyapunov analysis gives asymptotic gradient convergence and a linear rate under the additional PL condition. This continuous soft-polar transform differs from a fixed empirical polynomial with Frobenius normalization. Stochastic finite-precision Muon and arbitrary fixed practical schedules remain outside the result.

B.10 Generalized SOAP: rotation without covariance transport

Source. Sun et al. [13], arXiv:2609.24522v1, Algorithm 4; Theorem 1 in Section 3.3.1; proof in Appendix B. **Update.** Equation (46), with $M_0 = V_0 = 0$, arbitrary current-sample-dependent orthogonal rotations, and entrywise division and square root. **Schedule.** For $\gamma, \tau \in (0, 1]$, define $\widehat{\sigma}^2 = \max\{\sigma^2, L_F\Delta/(T\gamma^2)\}$. Require $\sqrt{V_{t,ij} + \varepsilon} \geq \delta > 0$ and choose

$$1 - \theta = \sqrt{\frac{L_F\Delta}{T\widehat{\sigma}^2}}, \quad 0 < \beta < 1, \quad \eta = \sqrt{\frac{\delta^2\Delta}{4L_F T\widehat{\sigma}^2}}, \quad \varepsilon = \frac{\tau\widehat{\sigma}^2}{m+n}. \tag{82}$$

Conclusion. Equation (47). Taking $\delta = \sqrt{\varepsilon}$ gives prefactor $(2 + 16/\sqrt{\tau})\sqrt{m+n}$. **Scope.** The result excludes decay, bias correction, transported second moments, and finite precision. The related SPlus theorem concerns truncated internal iterates, not the averaged output.

B.11 Two geometries: nonconvex adaptive families

Source. Xie et al. [11], arXiv:2511.20584v1, Algorithm 1, Theorems C.7–C.8, noise Assumption C.1, and normalized-descent comparisons in Theorems 4.6 and 4.9. **Guarantee and assumptions.** Cumulative and EMA adaptation over well-structured metric cones controls the average induced dual-gradient norm at $\widetilde{\mathcal{O}}(T^{-1/4})$, including one-sided Shampoo. Besides unbiasedness, Assumption C.1 requires, almost surely,

$$-\Sigma \preceq \nabla F(x)\nabla F(x)^\top - g(x)g(x)^\top \preceq \Sigma. \tag{83}$$

This is stronger than a noise covariance envelope. The EMA specialization uses $1-\beta = \Theta(\log d/T)$ and $\eta \asymp \sqrt{\Delta/(\Lambda_{\mathcal{H}} T \log^2 d)}$, with adaptive smoothness $\Lambda_{\mathcal{H}}$ and the structured noise trace in the constants. The NSD adaptive-noise comparison changes the noise premise, while the dimension-dependent lower bound concerns sign descent. Arbitrary two-sided updates, momentum/decay, and finite NS are outside the guarantee.

C Reference specifications and historical context

These tables expand section 4.5, recording the recursions, stored state, numerical work, and distributed implementations used in the comparison.

C.1 Representative method specifications

Tables 6 and 7 describe representative recursions for an $m \times n$ block. As in (9), D_t denotes the gradient-driven direction; learning-rate schedules, decoupled decay, and block scales are outer choices. The referenced equations specify initialization, damping, and averaging.

Table 6: State, update input, and geometry. KL population targets and SOAP fixed-basis equations are distinguished from their online updates.

Method / anchor	Retained state and input	Direction and represented geometry
AdamW [2]	Entrywise EMA states M_t, V_t, fed by G_t and $G_t^{\odot 2}$	Bias-corrected diagonal scaling of momentum, (8)
Cumulative Shampoo [3]	L_t, R_t: cumulative mode Grams; input G_t	$L_t^{-1/4} G_t R_t^{-1/4}$; one separable operator, (12)
EMA Shampoo family [54]	Momentum M_t and EMA mode Grams L_t, R_t, all fed by G_t	Damped one-/two-sided powers applied to M_t; the analyzed powers and decay are in (44)
SOAP [5]	Mode Grams/bases and entrywise moments in $Q_L^\top G_t Q_R$	Rotated Adam scaling, (15); the basis is factored but the scales need not be
KL-Shampoo [10]	Coupled factor estimates; practical bases and factor eigenvalue scales; gradient or momentum input	Inverse-square-root factor action. (18) specifies a population target; online estimation and the practical QR scheme are separate recursions (source Secs. 3.2–4)

Table 6 continued

Method / anchor	Retained state and input	Direction and represented geometry
ASGO, Algorithm 1 [18]	One left accumulator $V_t^L = \sum_{s=0}^{t} G_s G_s^\top$; input G_t	$[(V_t^L)^{1/2} + \varepsilon I]^{-1} G_t$; damping is outside the square root
Finite Muon [6, 31]	Momentum buffer; input Z_t with a declared momentum convention	Scaled output $a_t X_K$ of (23); a specified finite singular-value map, (24)

Table 7: Numerical and distributed implementations. State orders count dense unblocked optimizer arrays, excluding parameters, gradients, and transient workspace; constants depend on caching.

Method	State order	Numerical work / refresh	Requirement for the same logical update
AdamW	$O(mn)$	Elementwise state and scaling every step	Coordinate shards retain the corresponding moments and synchronized gradient input
Cumulative Shampoo	$O(m^2 + n^2)$	Current exact powers in the canonical recursion	Factor sharding preserves full mode Grams and their action
EMA Shampoo family	$O(m^2 + n^2 + mn)$	Current powers in the stated theorem; cached roots or blocking must be specified separately	Synchronize EMA/momentum states and preserve the chosen logical blocks
SOAP	$O(m^2 + n^2 + mn)$	Basis refresh by EVD or warm-started QR; coordinate moments each step	Preserve rotations, basis ownership, and the state rule at each basis change
KL-Shampoo	$O(m^2 + n^2 + mn)$ with momentum	Coupled online estimation; QR basis refresh and eigenvalue-scale updates in the practical scheme	Preserve weighted cross-mode products and the same estimator/refresh schedule
ASGO, Algorithm 1	$O(m^2)$	Current square root and damped inverse	Preserve the full selected-mode accumulator; a row-block diagonal substitute changes it
Finite Muon	$O(mn)$	K polynomial steps with specified normalization and precision	Preserve global normalization and products; proposition 8.2 gives one exact sharded construction

Dense arithmetic costs. AdamW requires $O(mn)$ elementwise work. Forming both mode Grams and applying dense two-sided factors or bases costs $O(mn(m+n))$, with $O(m^3 + n^3)$ for full EVD or QR refreshes. A left-only mode needs $O(m^2 n)$ products and a dense $O(m^3)$ decomposition. A K-step Muon polynomial using the smaller Gram costs $O(Kmn \min(m, n))$. Runtime also depends on refresh frequency, solver depth, blocking, kernel utilization, and communication; iterative KL fitting adds work according to its schedule.

The associated analyses differ for the same reason as the updates. SOAP enlarges the scaling class within a factored basis, raising questions about state transport. Finite Muon changes the singular response while preserving singular directions in exact arithmetic, motivating alignment and energy bounds. KL fitting changes what the factors estimate, whereas sharding can preserve an existing operator. Estimator tracking and distributed product fidelity address these last two choices, respectively.

C.2 Conceptual anchors

Table 8: Conceptual anchors for the comparison.

Anchor	Primary focus	Role in the present synthesis
Old Optimizer, New Norm [7]	Norm-based interpretations of instantaneous first-order updates	Separate direction, magnitude, and the effects of historical adaptation
Clarifying Shampoo [9]	Adapted spectral descent and trajectory-dependent Shampoo geometry	State conditions for adapted-polar identities and distinguish them from identical dynamics
KL-Shampoo / KL-SOAP [10]	Divergence-based factor estimation and practical basis/scale updates	Separate a population fitting target, its online estimator, and the resulting operator
A Tale of Two Geometries [11]	Adaptive versus non-Euclidean smoothness and stochastic variance	Compare rates with their geometry and oracle assumptions retained
ARO [12]	Adaptive rotation, normed descent, and matrix optimizer design	Place moving-basis adaptation and spectral transformations in a shared specification
Rotation-based convergence [13]	A unified analysis of specified SOAP-, Conda-, and SPlus-type recursions	Attach rotation-based claims to the analyzed normalizer, state, and schedules

C.3 Historical development

Table 9: Development of structured preconditioning and spectral update design.

Period	Structured adaptive preconditioning	Spectral update design and Muon
2018–2023	Original Shampoo and its cumulative regret analysis; scalable and distributed realizations [3, 4, 26]	The later Muon line is not yet present in this chronology; norm-based and structured-preconditioning ancestry should not be conflated.
2024	Statistical reinterpretation of the factors; SOAP separates a learned basis from coordinate scaling [5, 20]	Norm-based synthesis and Muon's momentum-to-polar construction, computed through matrix polynomials [6, 7]

Table 9 continued

Period	Structured adaptive preconditioning	Spectral update design and Muon
2025	Heuristic decomposition, stable whitening, KL fitting, and structure-sensitive convex guarantees [10, 18, 19, 21, 25]	Scaling recipes, adaptive spectral updates, and communication-aware compression [8, 40, 41, 45]
2026	Practical-form nonconvex theory, adapted-polar connections, rotation-based convergence, and solver/precision-aware refinements [9, 13, 28, 29, 54]	Taylor-NS and actual finite-map guarantees; rotation-based design, soft proxies, and curvature-aware compositions [12, 31–33, 42, 52]

D Empirical records

The six observations in section 9 are recorded below with source versions, locations, numerical settings, and the status of fields not established in this survey. Unverified fields are left explicit rather than inferred from another experiment in the same paper.

Table 10: Empirical observations. NE means *not extracted for this observation*, including fields whose applicability remains to be checked, rather than omitted by the paper. Precision refers only to the named computation.

Source / locator	Setting, budget, tuning, and numerical recipe	Metric and observed outcome
SOAP v2 [5] Fig. 1 (right), Secs. 5, 6.1–6.2; App. A	C4; 660M decoder-only model (non-embedding count); reference regime: 20× model-size tokens, 2M-token batches. Learning-rate sweep, then paired sweeps of learning rate and each other hyperparameter. Refresh intervals from 1 to 100; SOAP uses initialized eigenvectors and subsequent QR-based updates. Mixed-bf16 training; factor updates in fp32.	Loss versus refresh interval. SOAP and Shampoo are close at interval 1; both degrade as the interval increases, with slower degradation for SOAP. This is evidence for different basis and scale adaptation timescales in the evaluated regime.
Purifying Shampoo v1 [21] Fig. 2, Sec. 3; App. E.1	Imagewoof; SimpleViT and ConvNeXt V2; 100 epochs, batch 128. Learning-rate grids differ by method/model; $\epsilon = 10^{-10}$. Grafted Shampoo and EShampoo refresh roots/bases every 100 steps, respectively; eigenvalue correction updates each step. One A100 per run. Precision: NE.	Final training cross-entropy versus learning rate. EShampoo matches or improves on Adam-grafted Shampoo for both models. Ungrafted Shampoo matches the ViT result but not ConvNeXt V2. The observation separates scale correction from model-dependent effects.
KL-Shampoo v10 [10] Fig. 9; Sec. 2 and App. H	450M Llama3. Random search: 120 runs per method. QR every 10 iterations in fp32; refresh frequency is explicitly not tuned for runtime. Data, token budget, batch size, other state precision, and search ranges specific to this panel: NE.	Progress per iteration and per wall-clock time. KL-Shampoo improves on Muon per iteration, while Muon is better per time; both improve on SOAP. The two denominators give different conclusions under this numerical schedule.
Scalable Muon v1 [8] Table 1, Sec. 3.1; recipe in Sec. 2.2	Approximately 800M dense model, with a two-layer $[H, 4H]$ MLP replacing SwiGLU. Evaluated at 4B tokens of a 20B-token schedule. Compares baseline scaling, RMS normalization, and shape-adjusted learning rate. Recipe: five Newton–Schulz steps, momentum 0.95. Panel-specific tuning budget, data, and precision: NE.	Validation loss: baseline 2.812; normalized update 2.789; adjusted learning rate 2.789. Weight-RMS diagnostics accompany the loss. This supports shape-aware scaling within the studied recipe.

Table 10 continued

Source / locator	Setting, budget, tuning, and numerical recipe	Metric and observed outcome
Polar Express v5 [35] Fig. 5, Secs. 4.2–4.3	GPT-2-Small (124M), FineWeb, 1B tokens; batch 32, four H100s. Fixed learning rate 0.05 and weight decay 0.1 in this ablation. Varies inner depth from 2 to 30 and includes exact SVD. Mixed-bf16 training; polynomial matrix-sign computation in bf16. SVD arithmetic: NE.	Final validation loss and training-step runtime. Five or six polynomial steps improve on two or three; further accuracy, including SVD, gives no loss improvement. SVD roughly doubles training-step runtime. The denominator includes forward/backward computation.
DASH v1 [29] Table 1, Sec. 5	Llama-953M, C4; 9,089 steps, 2M-token batch, sequence length 1,024; three seeds. AdamW learning rate selected from six candidates, then transferred through Adam grafting. Matched row: block 2,048, coupled Newton inverse-root solver, refresh every step, Frobenius normalization, fp32 inverse roots.	Mean validation perplexity and *optimizer-step* latency. DIST and DASH both give perplexity 11.87; latency is 675 versus 221 ms. This establishes a computation saving for the matched row; its timing excludes forward/backward passes.